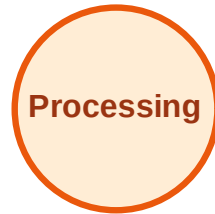


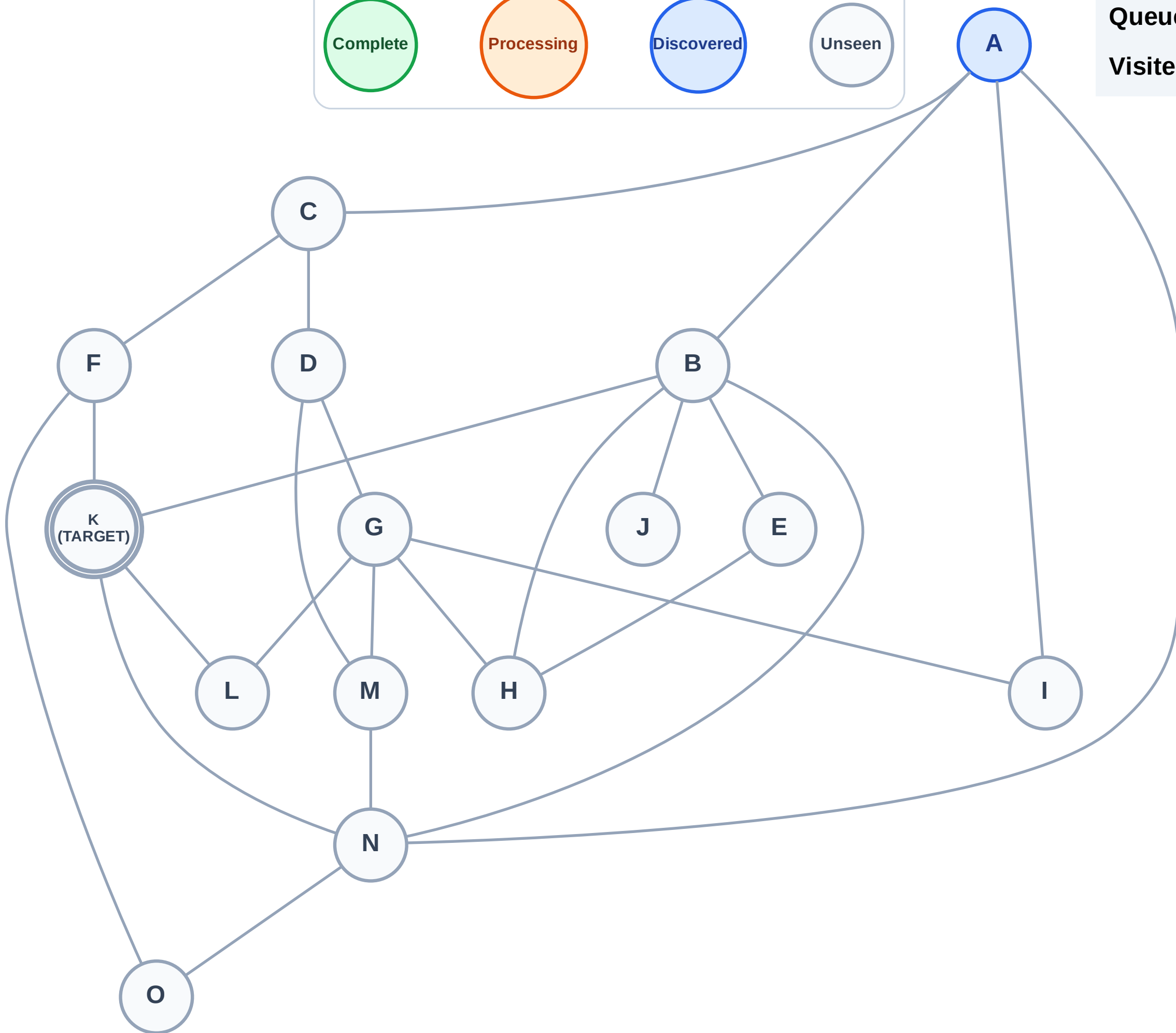
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



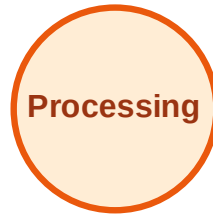
Queue: A
Visited: (none)



BFS Traversal

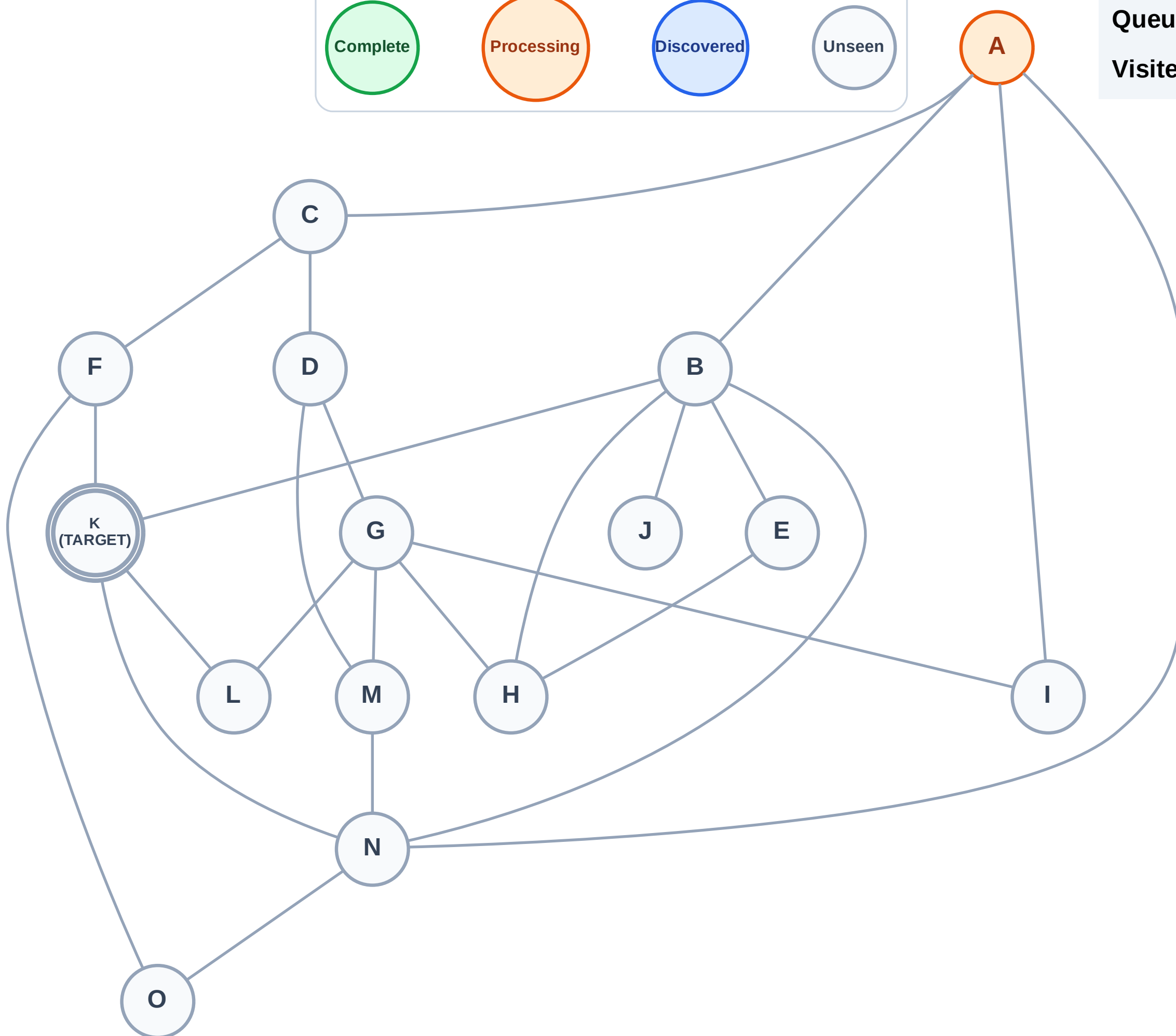
Step 2: Process node A

Legend



Queue: (empty)

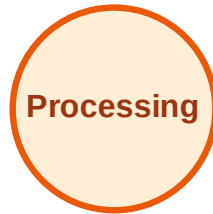
Visited: (none)



BFS Traversal

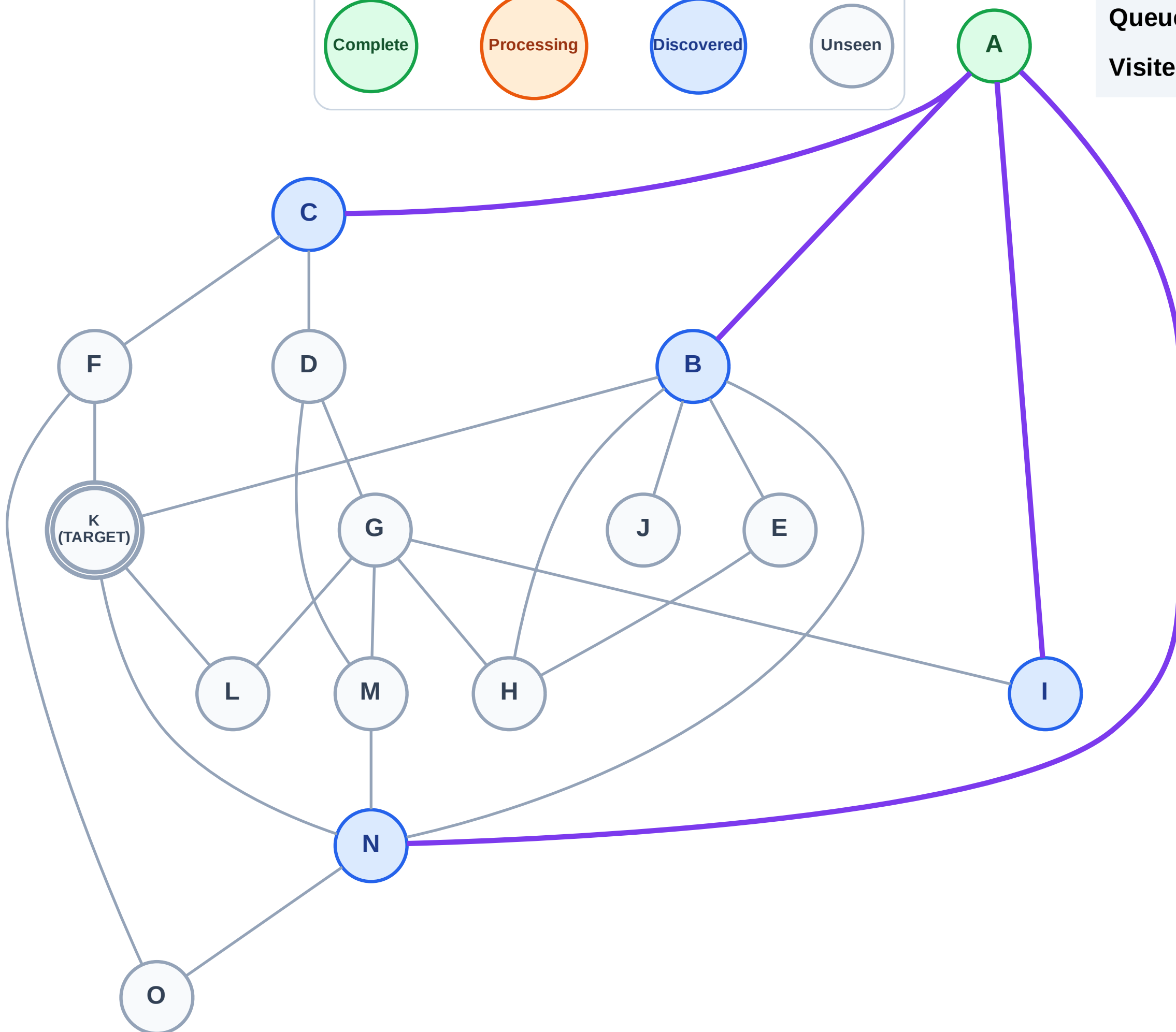
Step 3: Discover B, C, I, N from A

Legend



Queue: B → C → I → N

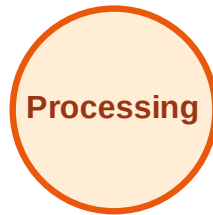
Visited: A



BFS Traversal

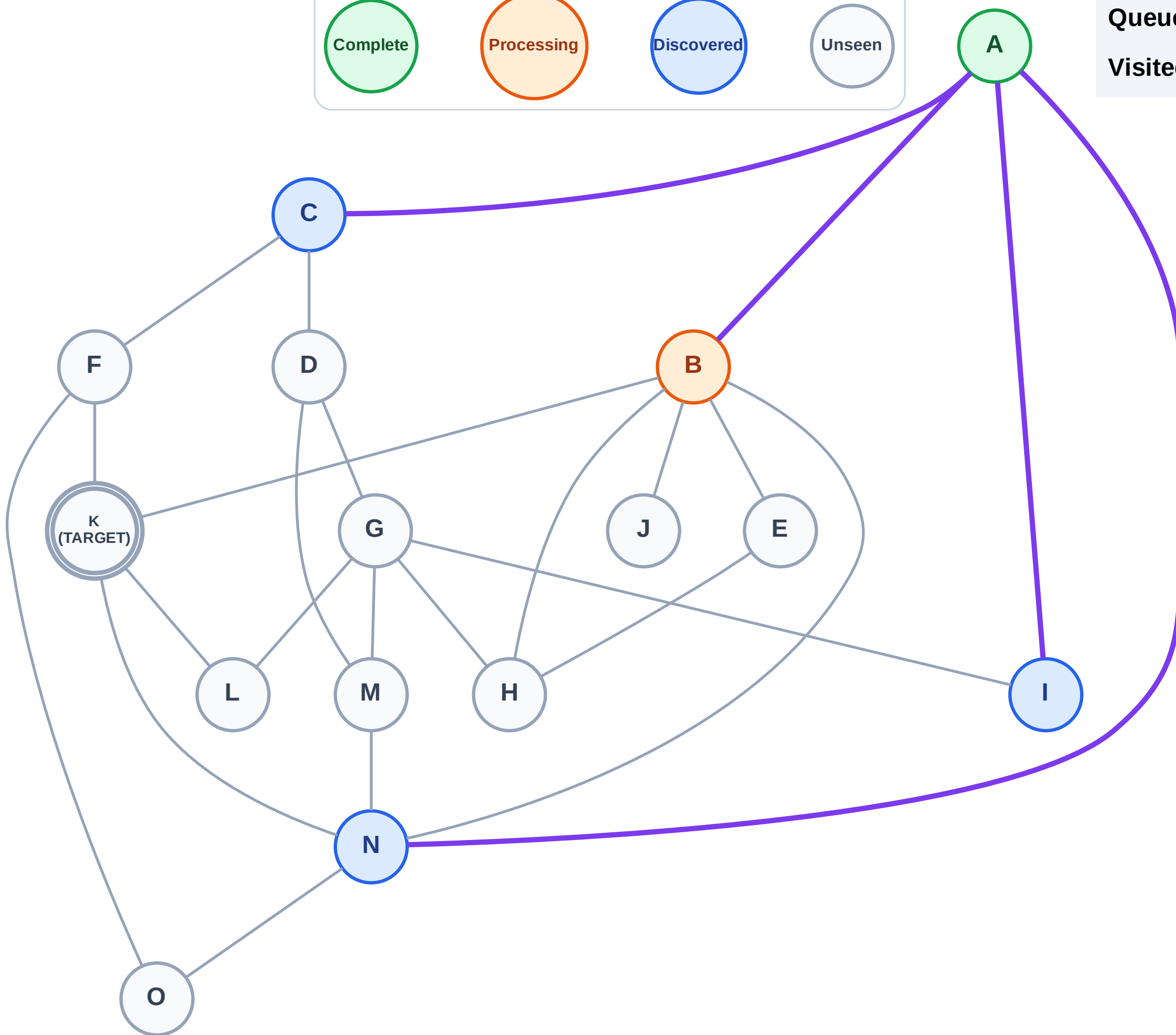
Step 4: Process node B

Legend



Queue: C → I → N

Visited: A



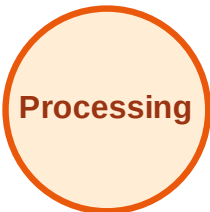
BFS Traversal

Step 5: Discover E, H, J, K from B

Legend



Complete



Processing



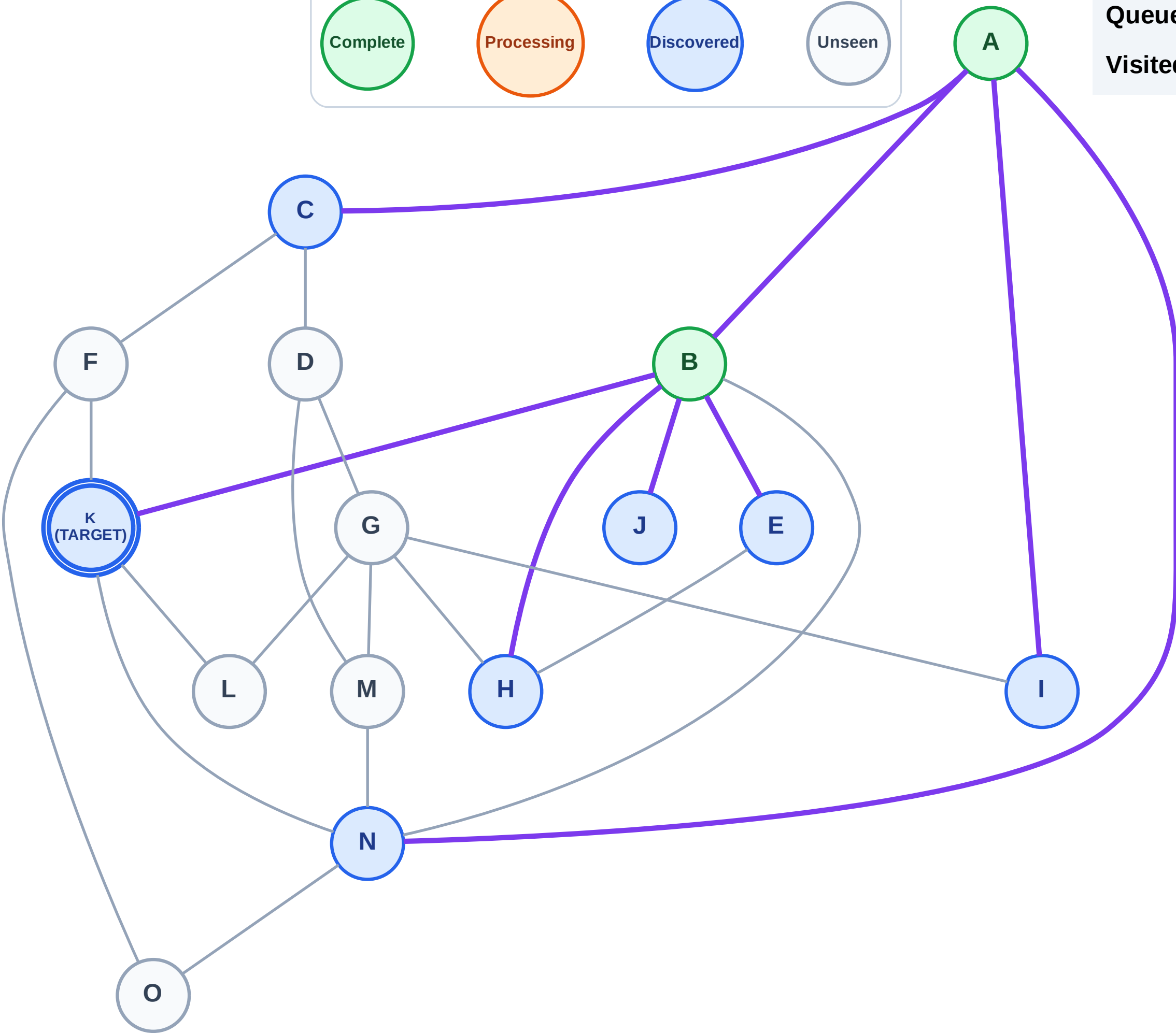
Discovered



Unseen

Queue: C → I → N → E → H → J → K

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

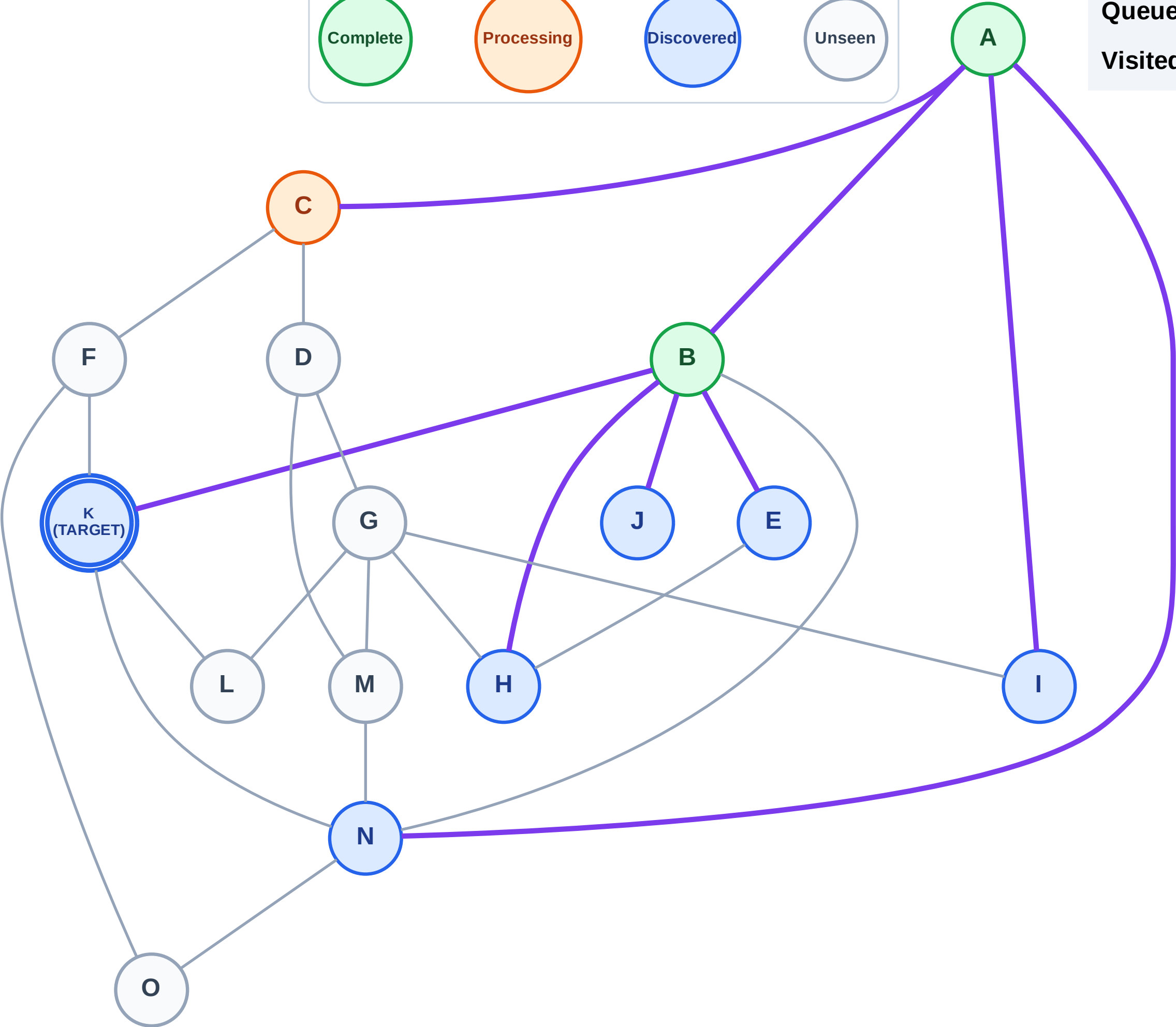
Processing

Discovered

Unseen

Queue: I → N → E → H → J → K

Visited: A, B



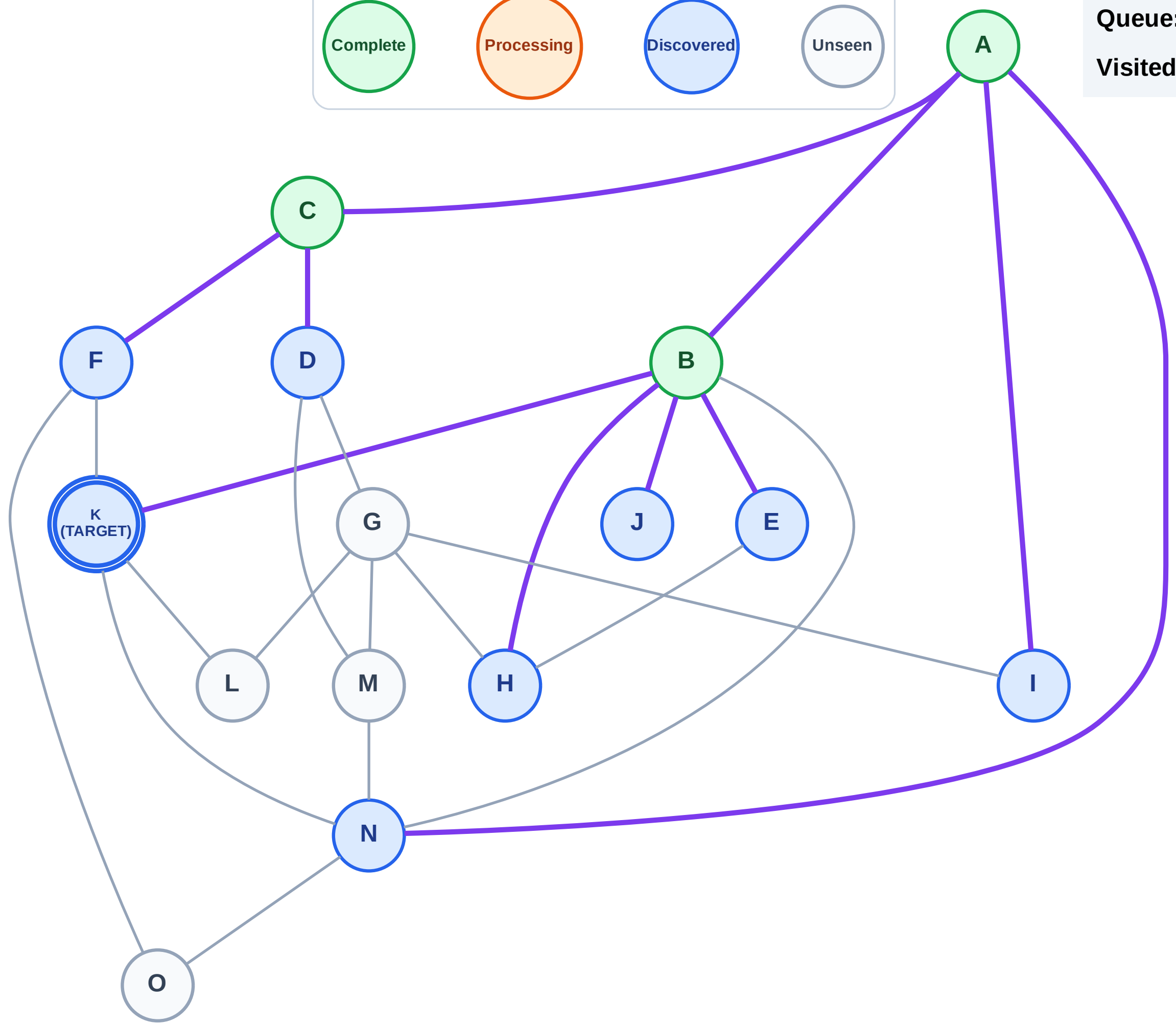
BFS Traversal

Step 7: Discover D, F from C

Legend



Queue: I → N → E → H → J → K → D → F
Visited: A, B, C



BFS Traversal

Step 8: Process node I

Legend

Complete

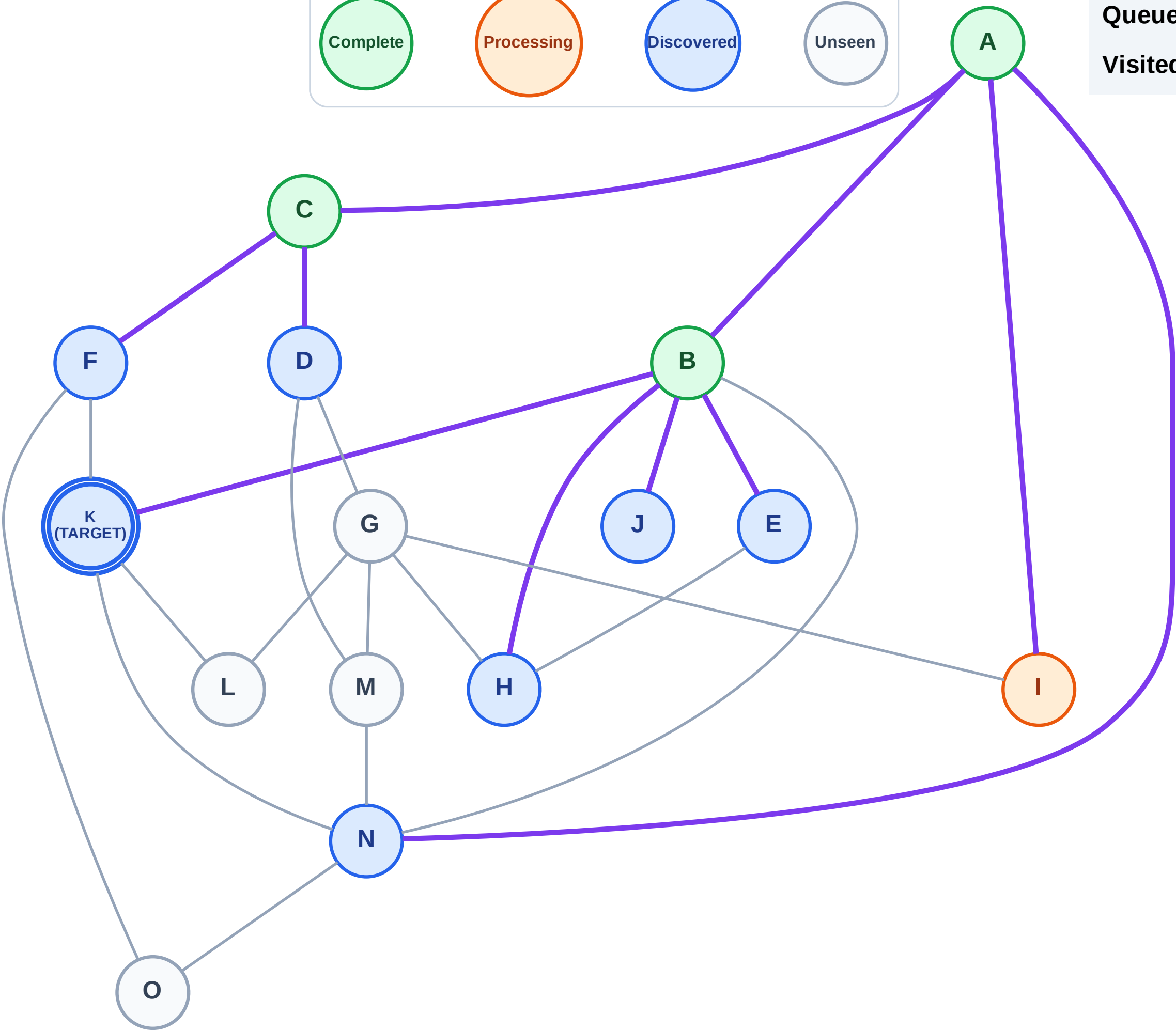
Processing

Discovered

Unseen

Queue: N → E → H → J → K → D → F

Visited: A, B, C



BFS Traversal

Step 9: Discover G from I

Legend



Complete



Processing



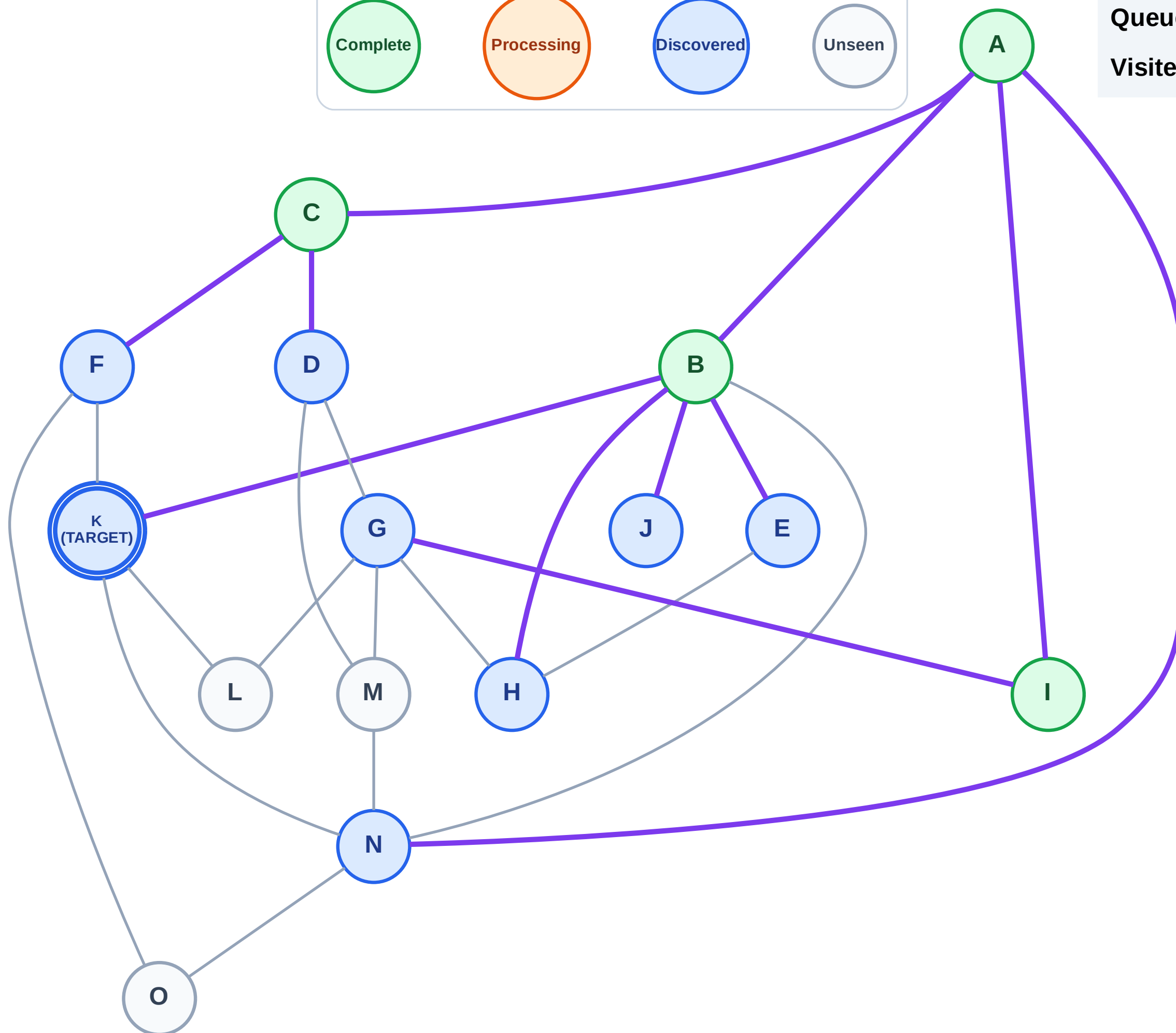
Discovered



Unseen

Queue: N → E → H → J → K → D → F → G

Visited: A, B, C, I



BFS Traversal

Step 10: Process node N

Legend

Complete

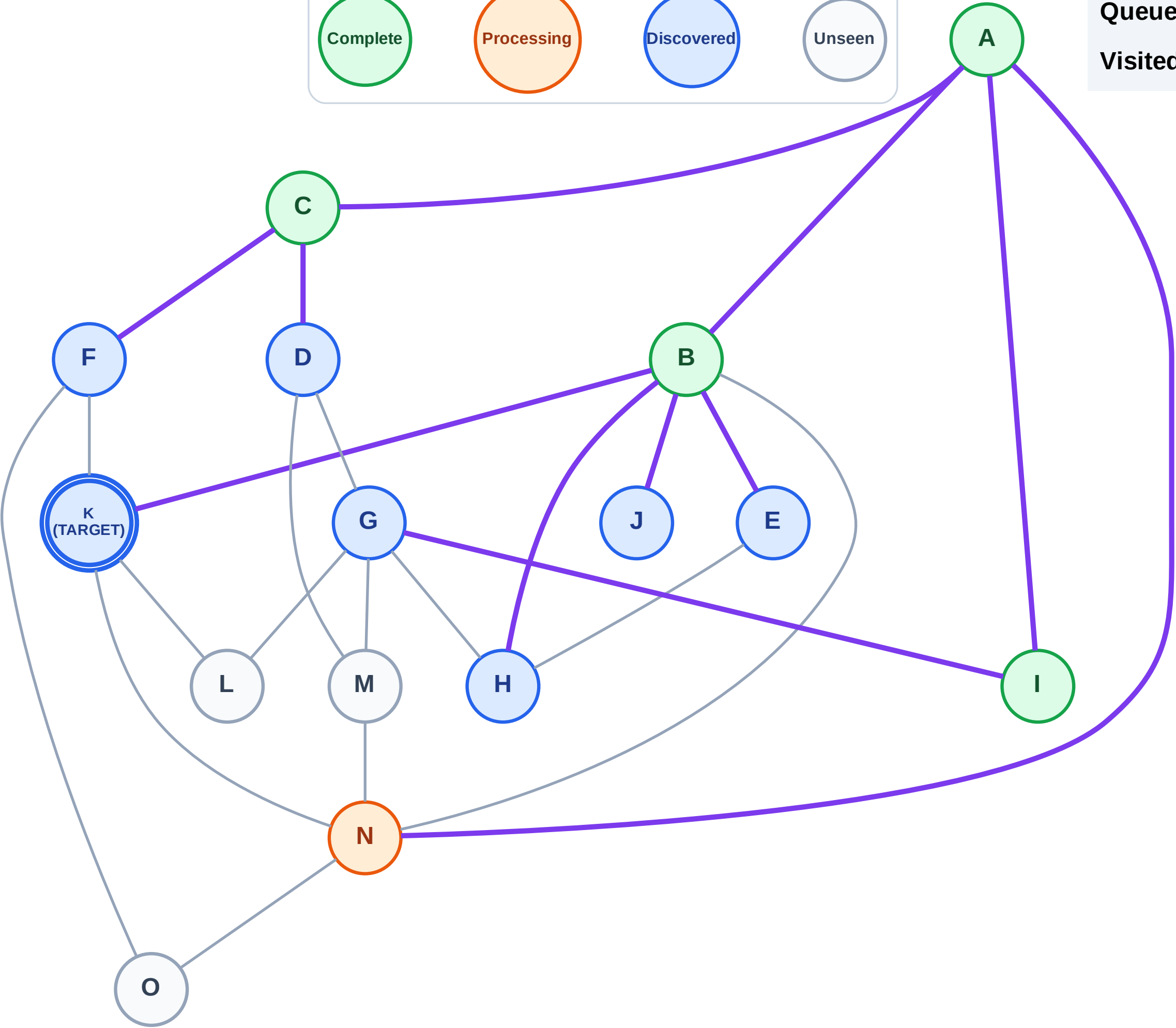
Processing

Discovered

Unseen

Queue: E → H → J → K → D → F → G

Visited: A, B, C, I



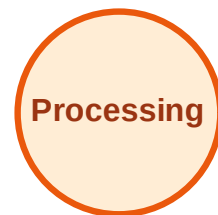
BFS Traversal

Step 11: Discover M, O from N

Legend



Complete



Processing



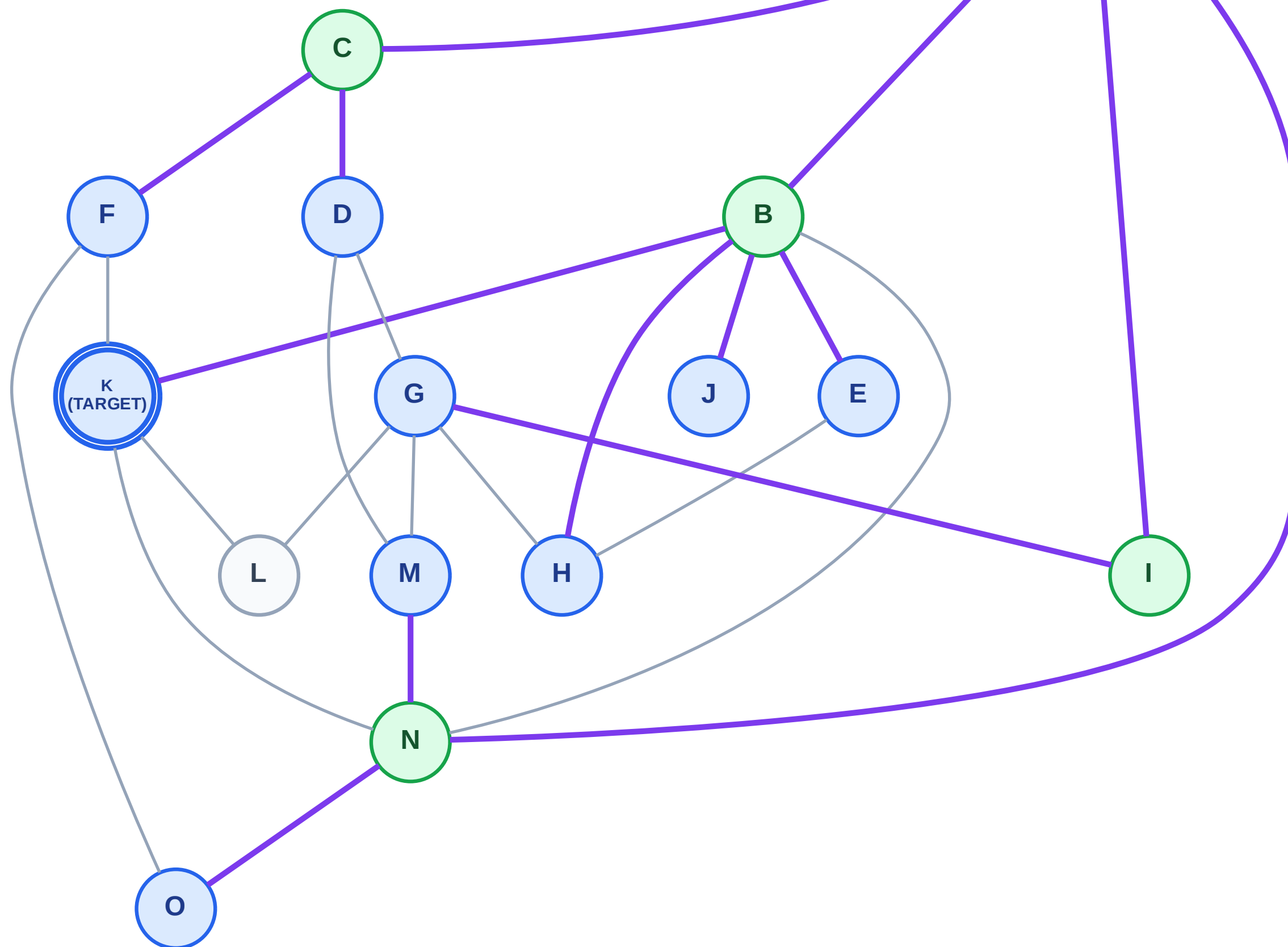
Discovered



Unseen

Queue: E → H → J → K → D → F → G → M → O

Visited: A, B, C, I, N



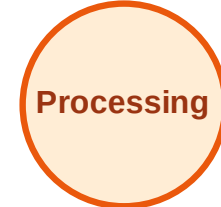
BFS Traversal

Step 12: Process node E

Legend



Complete



Processing



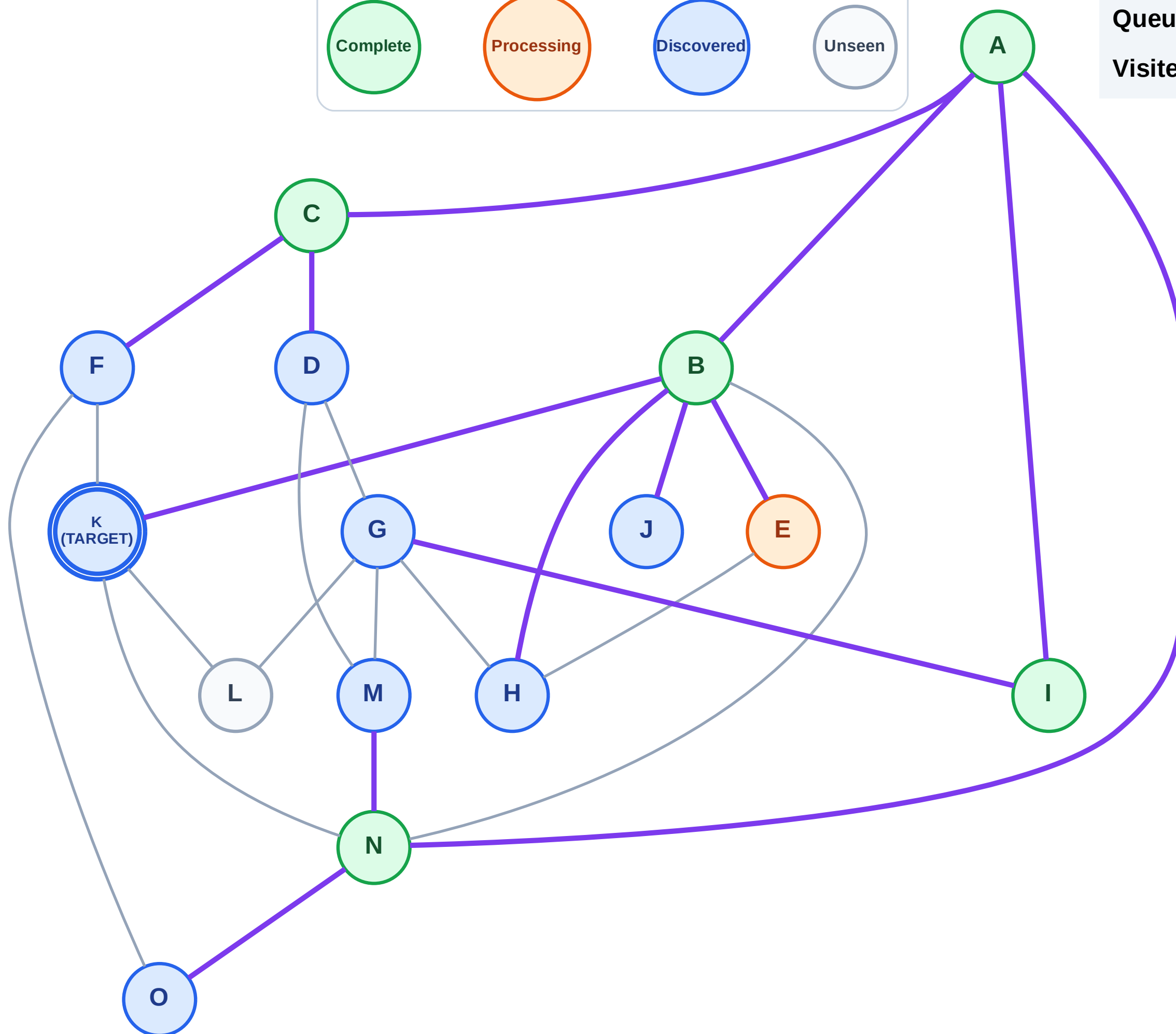
Discovered



Unseen

Queue: H → J → K → D → F → G → M → O

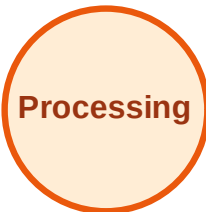
Visited: A, B, C, I, N



BFS Traversal

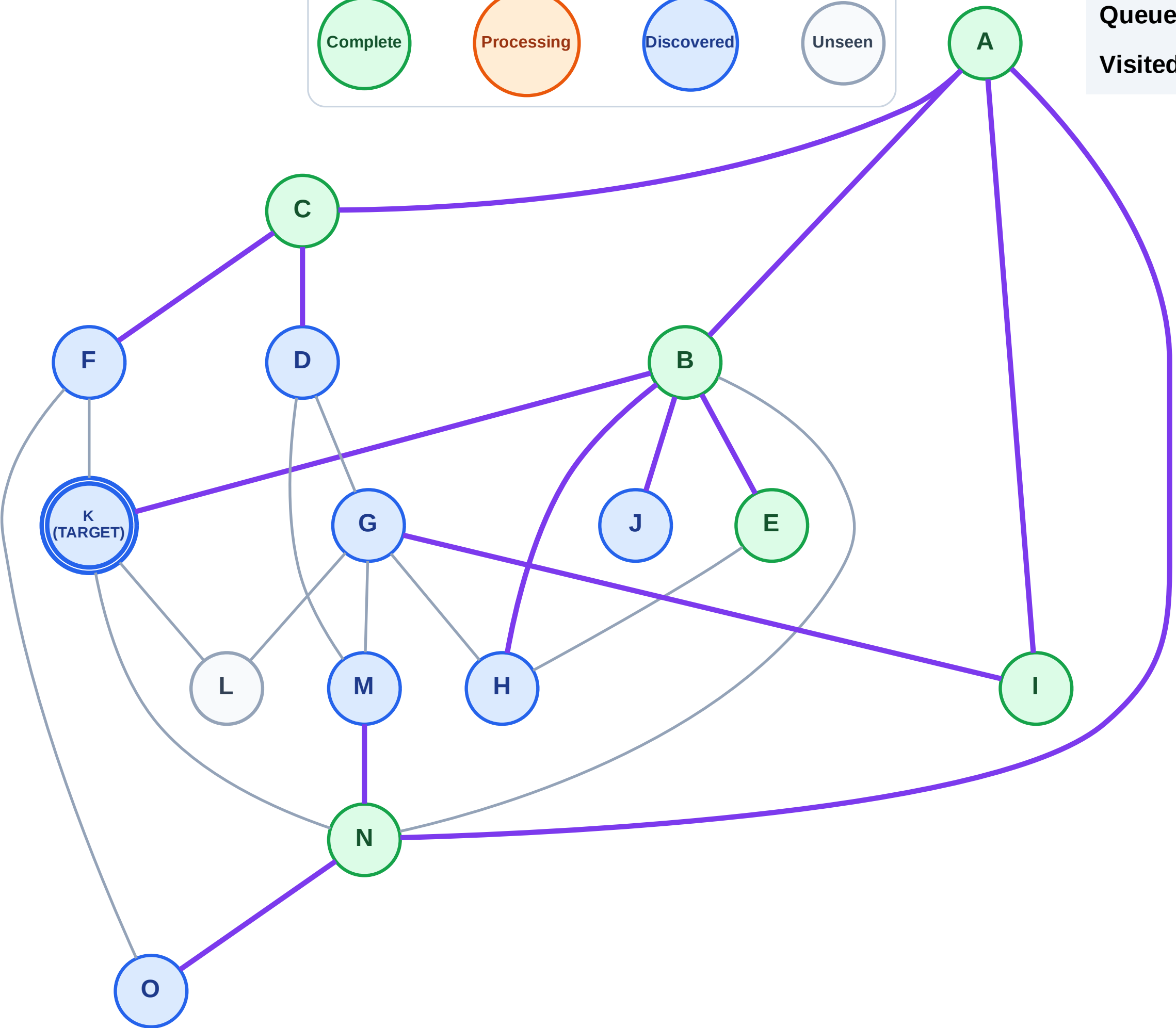
Step 13: No new neighbors from E

Legend



Queue: H → J → K → D → F → G → M → O

Visited: A, B, C, E, I, N



BFS Traversal

Step 14: Process node H

Legend

Complete

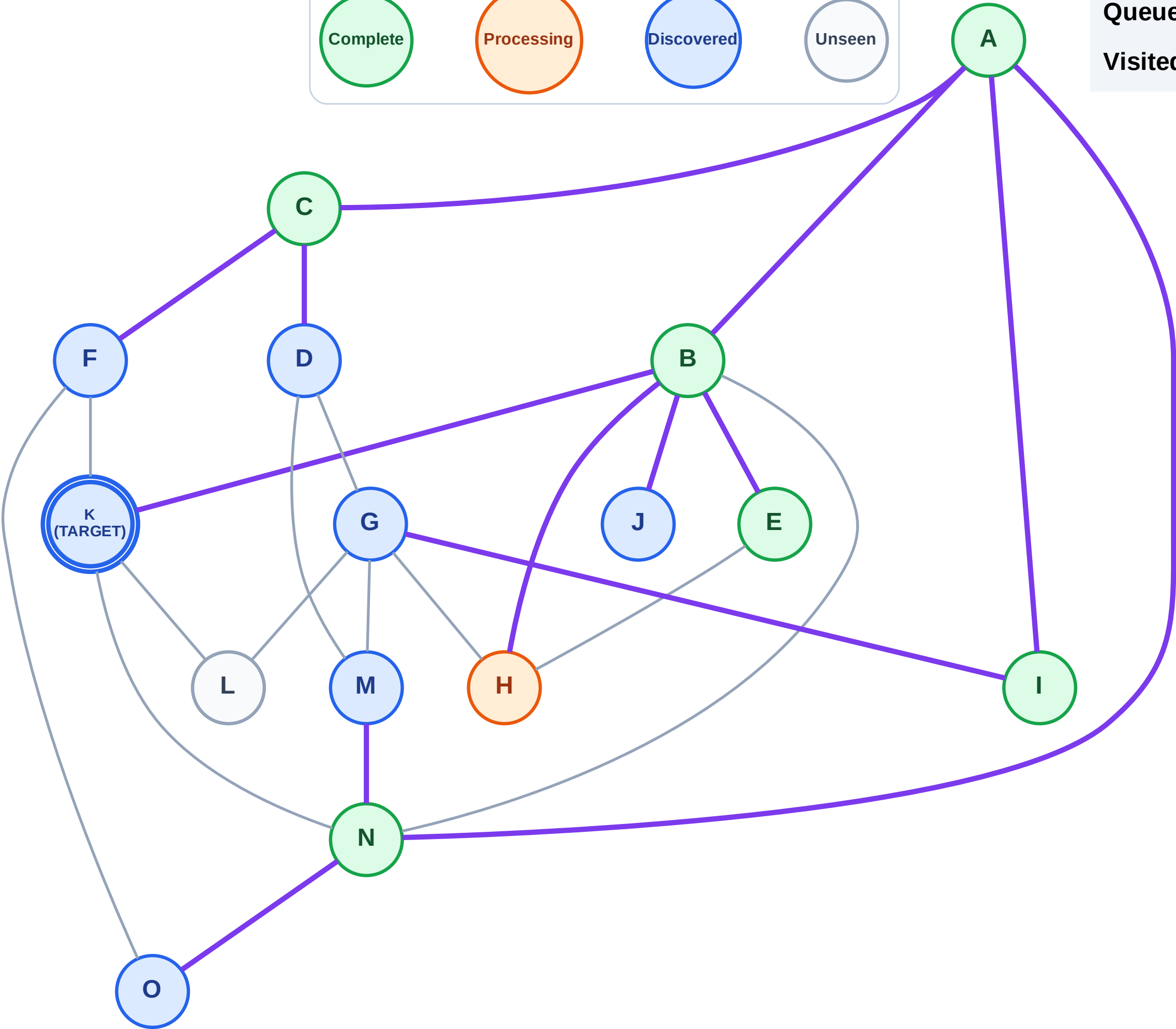
Processing

Discovered

Unseen

Queue: J → K → D → F → G → M → O

Visited: A, B, C, E, I, N



BFS Traversal

Step 15: No new neighbors from H

Legend

Complete

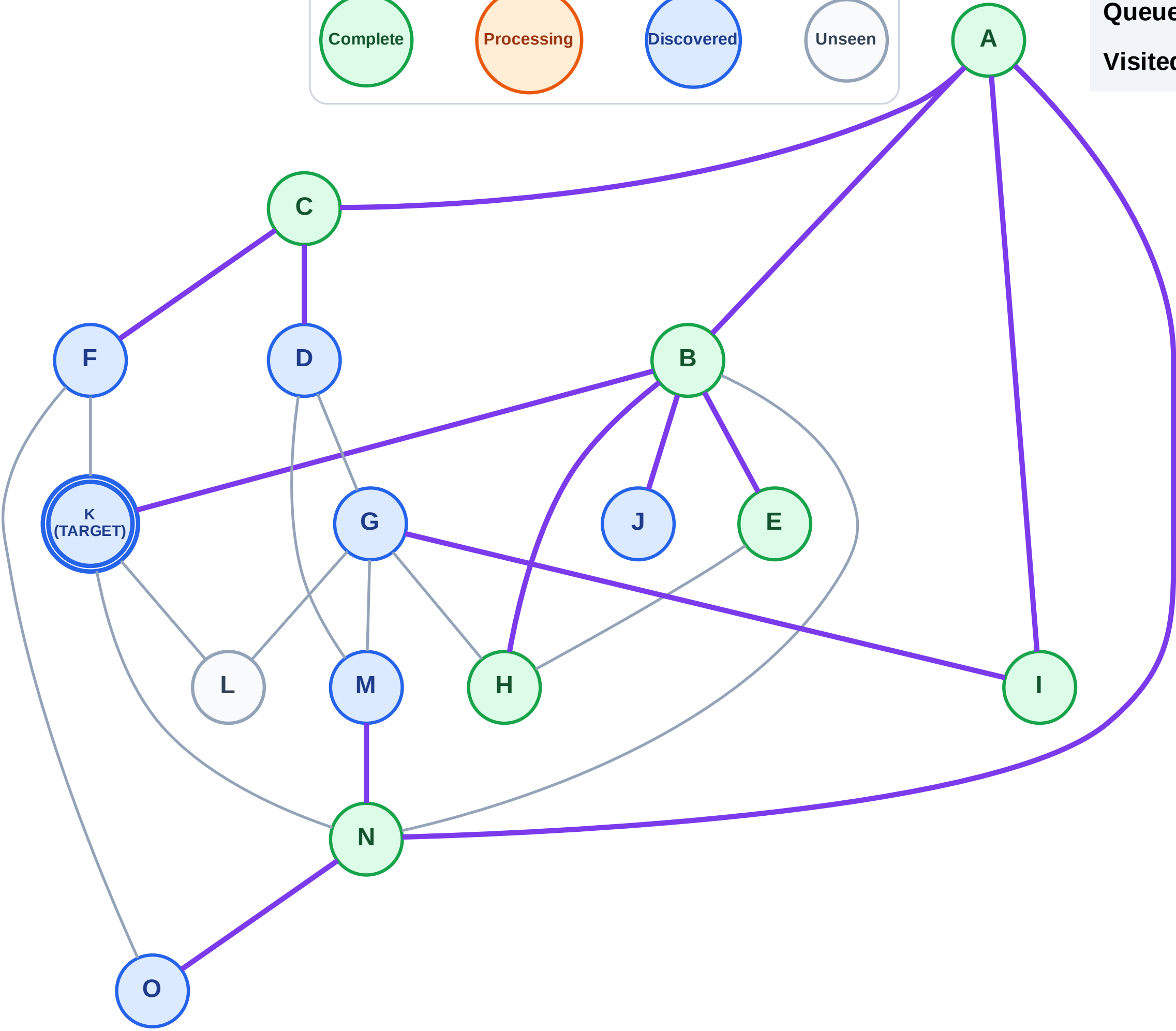
Processing

Discovered

Unseen

Queue: J → K → D → F → G → M → O

Visited: A, B, C, E, H, I, N



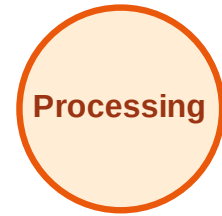
BFS Traversal

Step 16: Process node J

Legend



Complete



Processing



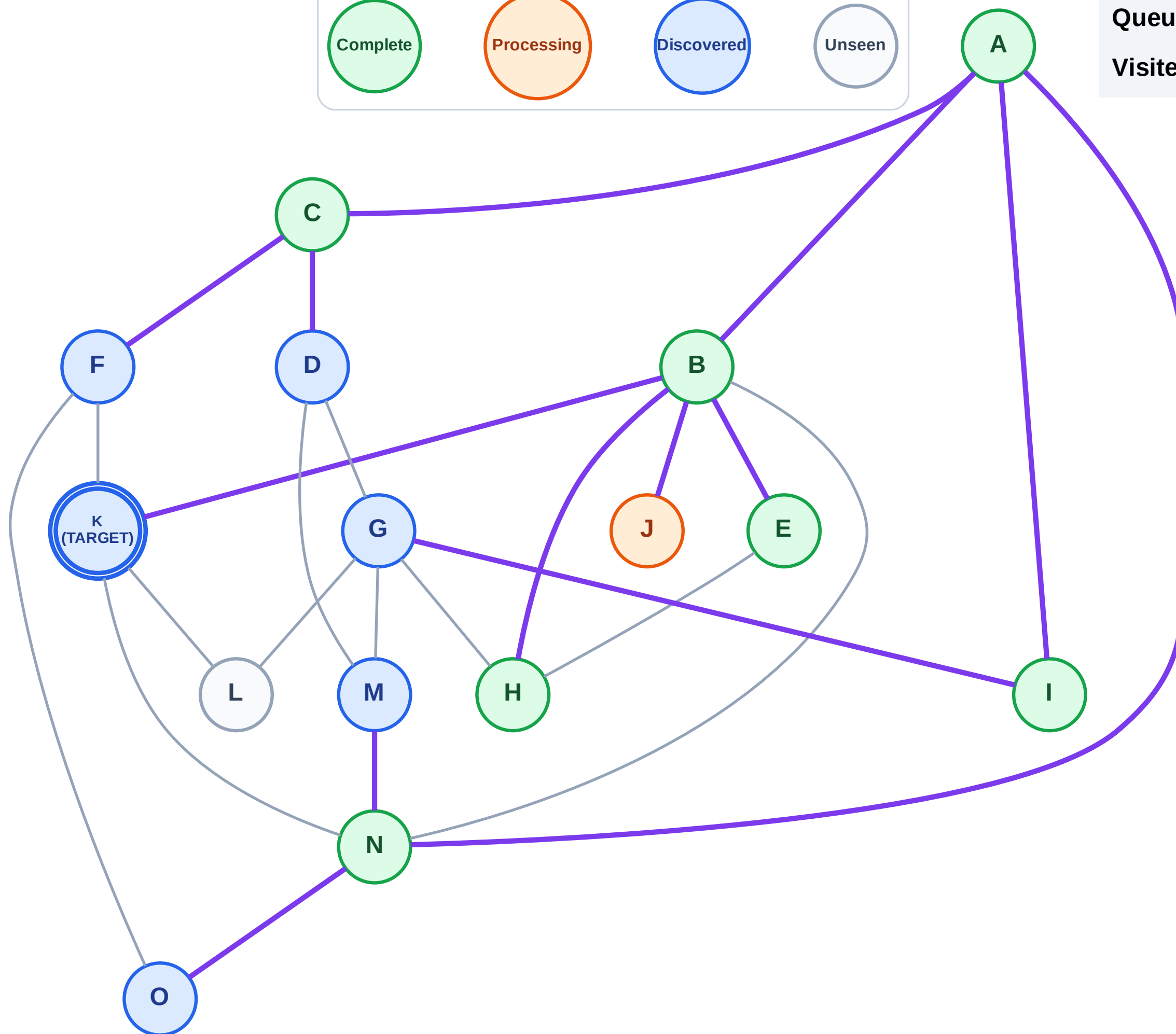
Discovered



Unseen

Queue: K → D → F → G → M → O

Visited: A, B, C, E, H, I, N



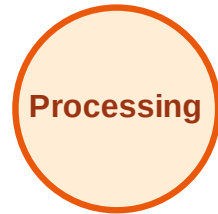
BFS Traversal

Step 17: No new neighbors from J

Legend



Complete



Processing



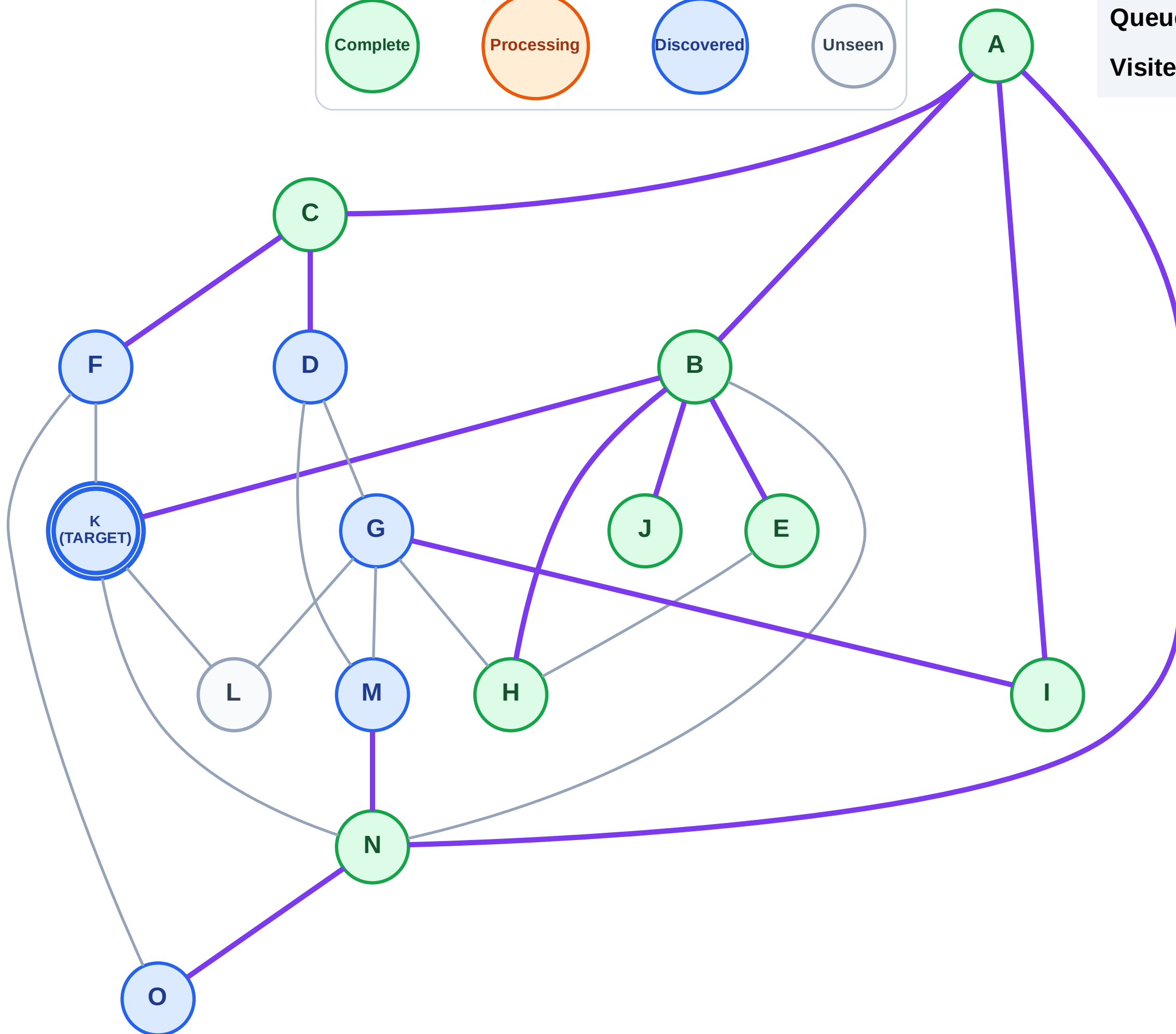
Discovered



Unseen

Queue: K → D → F → G → M → O

Visited: A, B, C, E, H, I, J, N



BFS Traversal

Step 18: Process node K

Legend

Complete

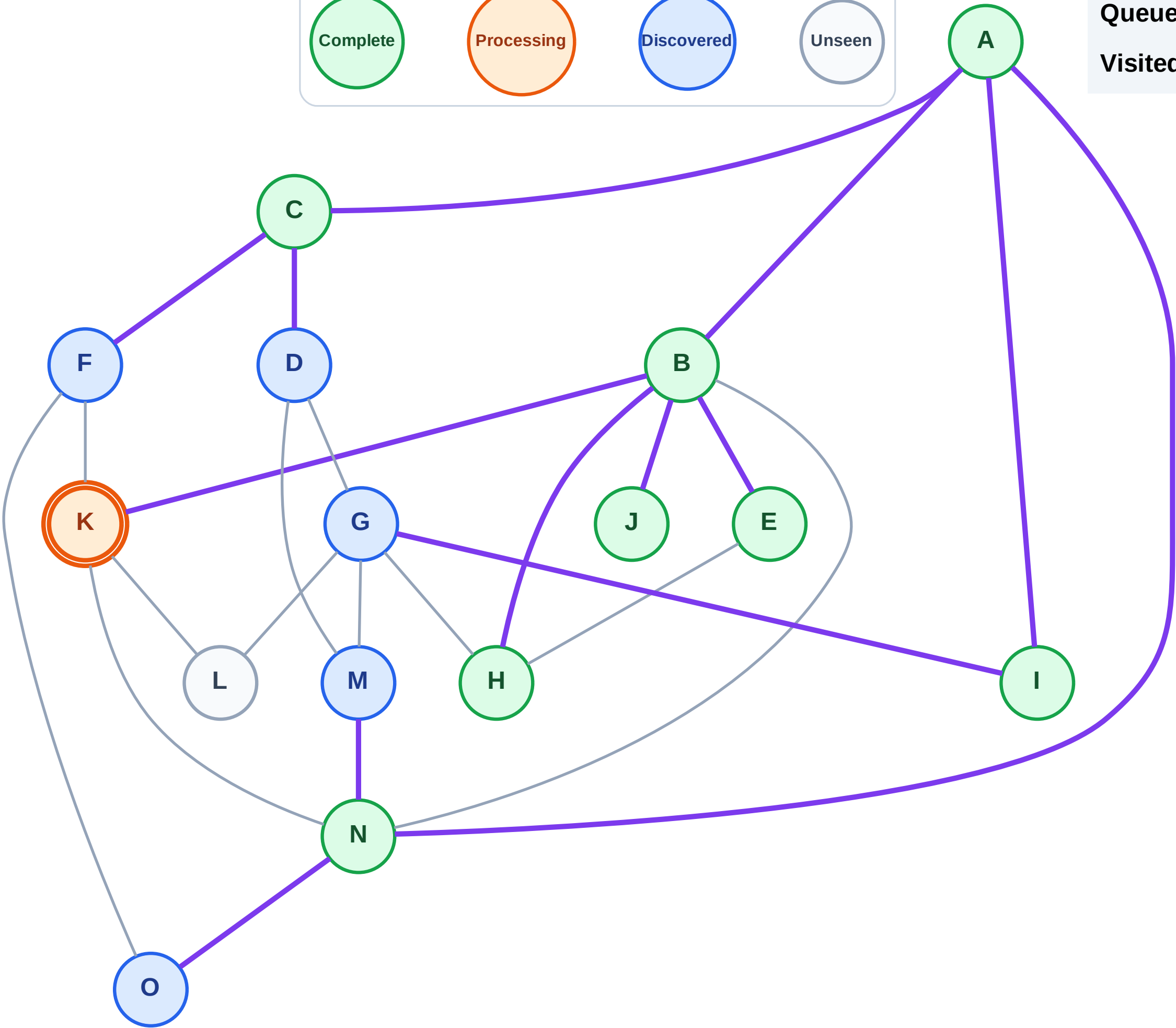
Processing

Discovered

Unseen

Queue: D → F → G → M → O

Visited: A, B, C, E, H, I, J, N



BFS Traversal

Step 19: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

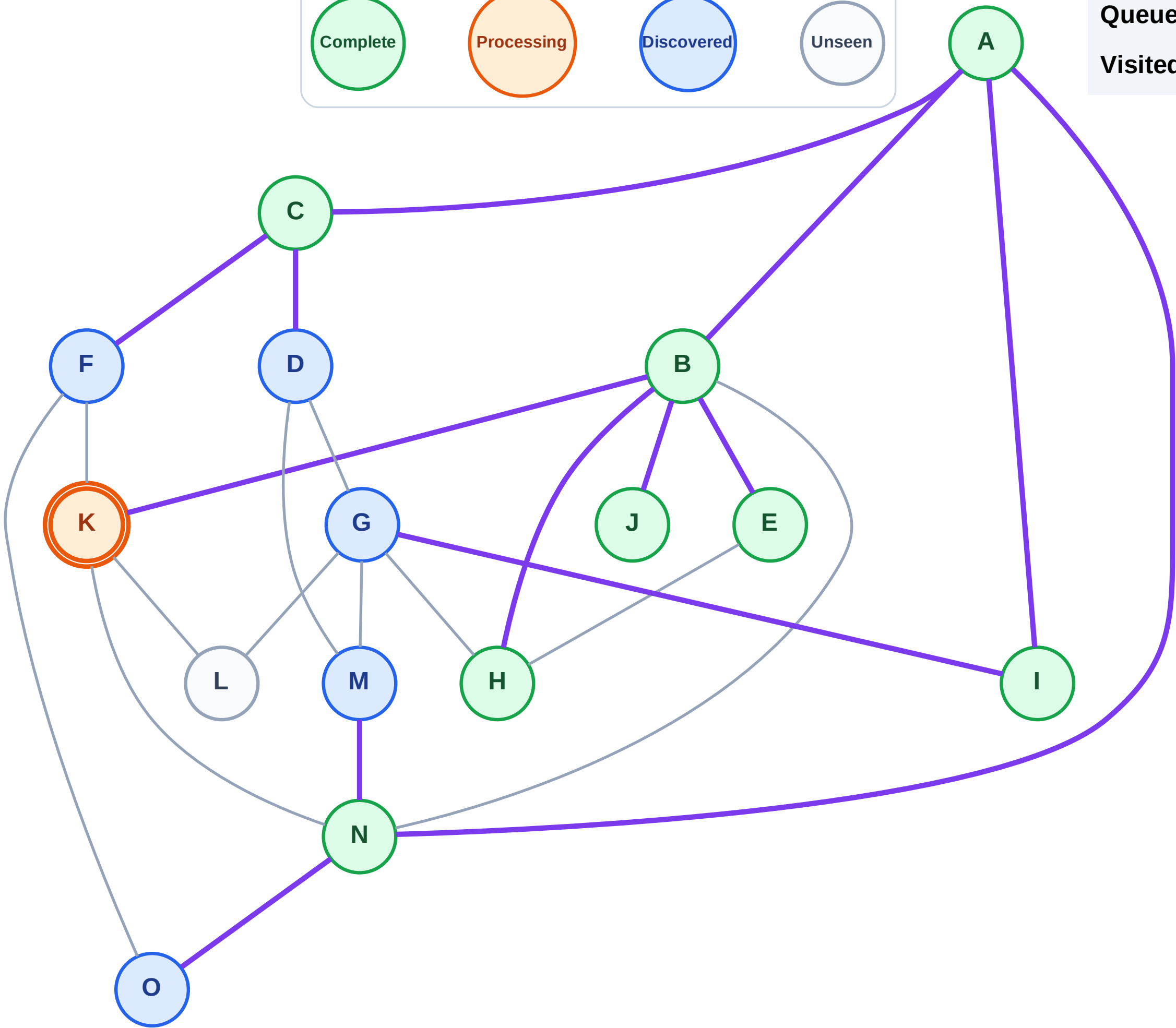
Processing

Discovered

Unseen

Queue: D → F → G → M → O

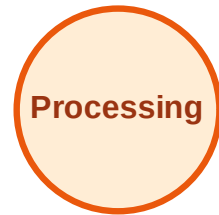
Visited: A, B, C, E, H, I, J, K, N



DFS Traversal

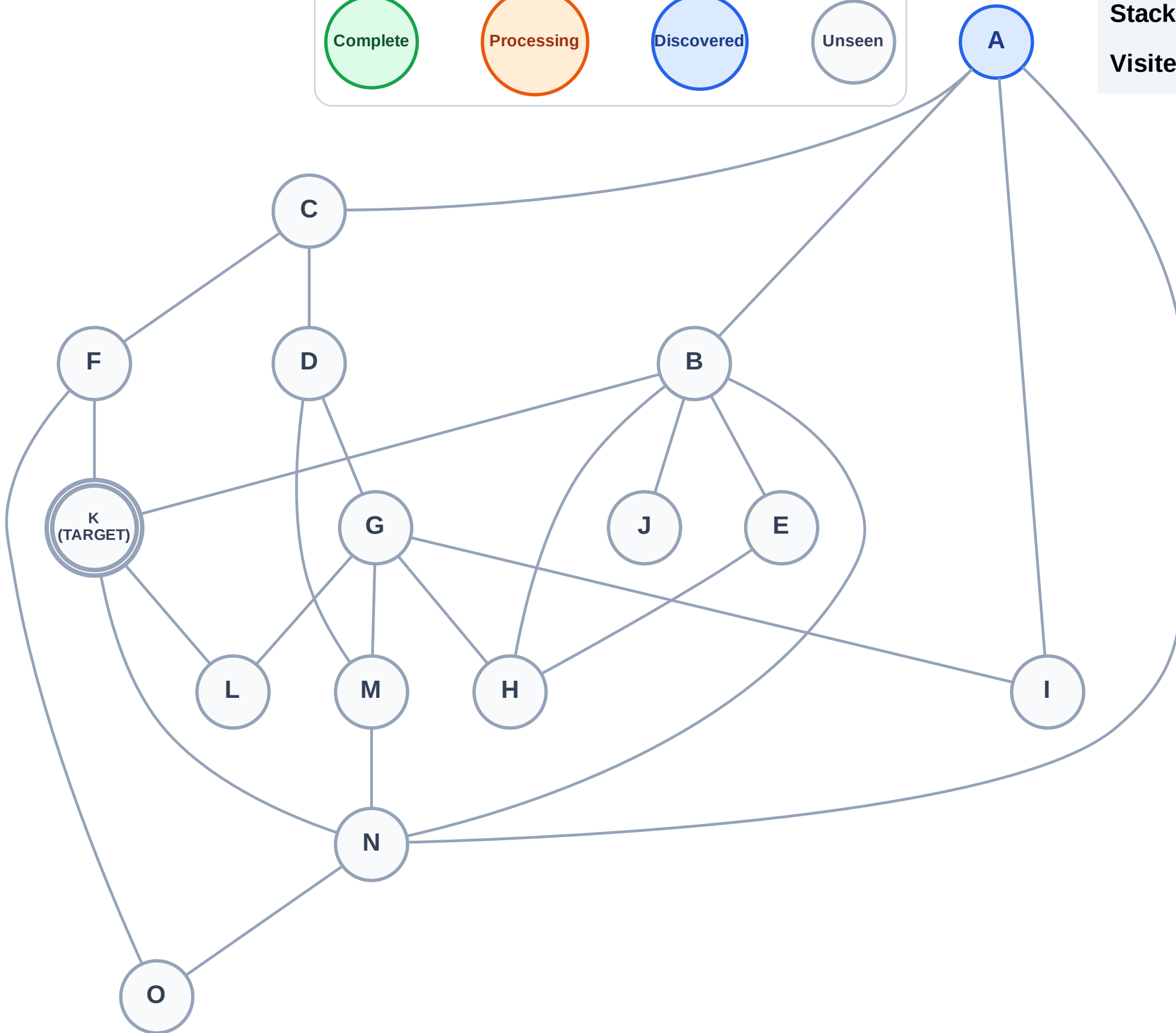
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

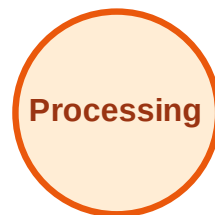
Visited: (none)



DFS Traversal

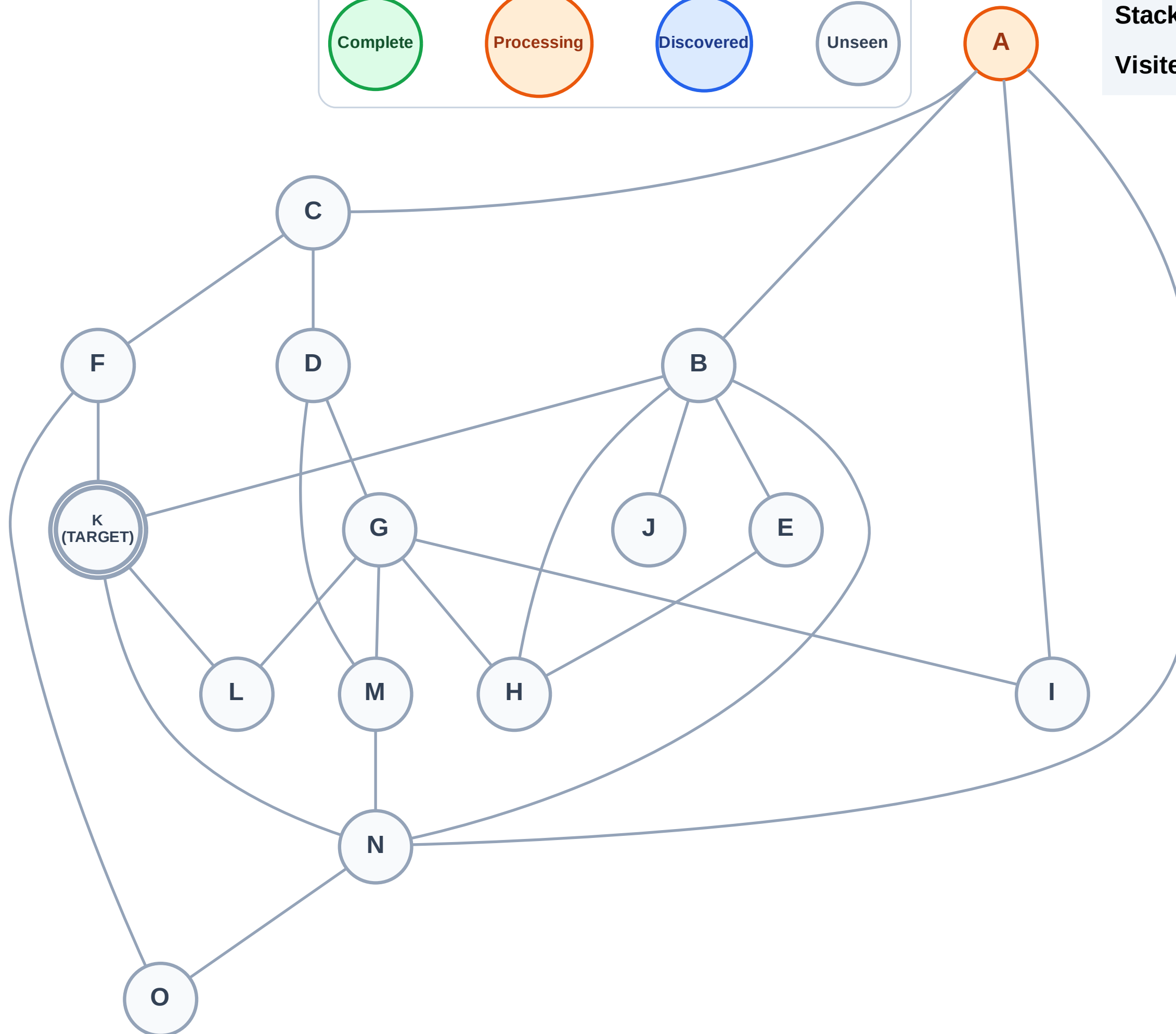
Step 2: Process node A

Legend



Stack (top at right): (empty)

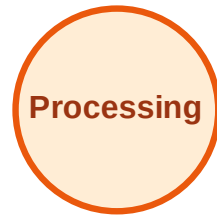
Visited: (none)



DFS Traversal

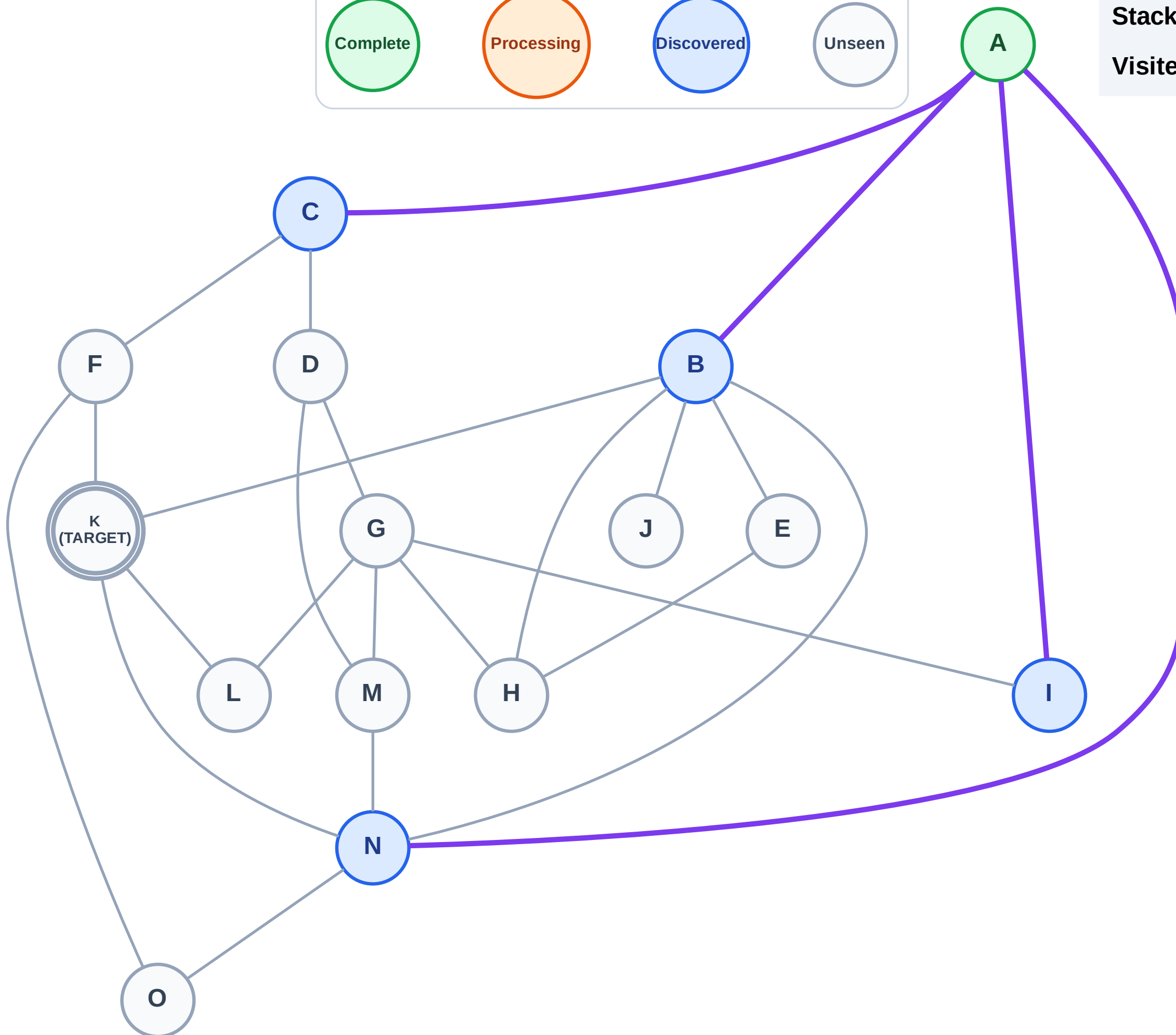
Step 3: Discover N, I, C, B from A

Legend



Stack (top at right): N | I | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

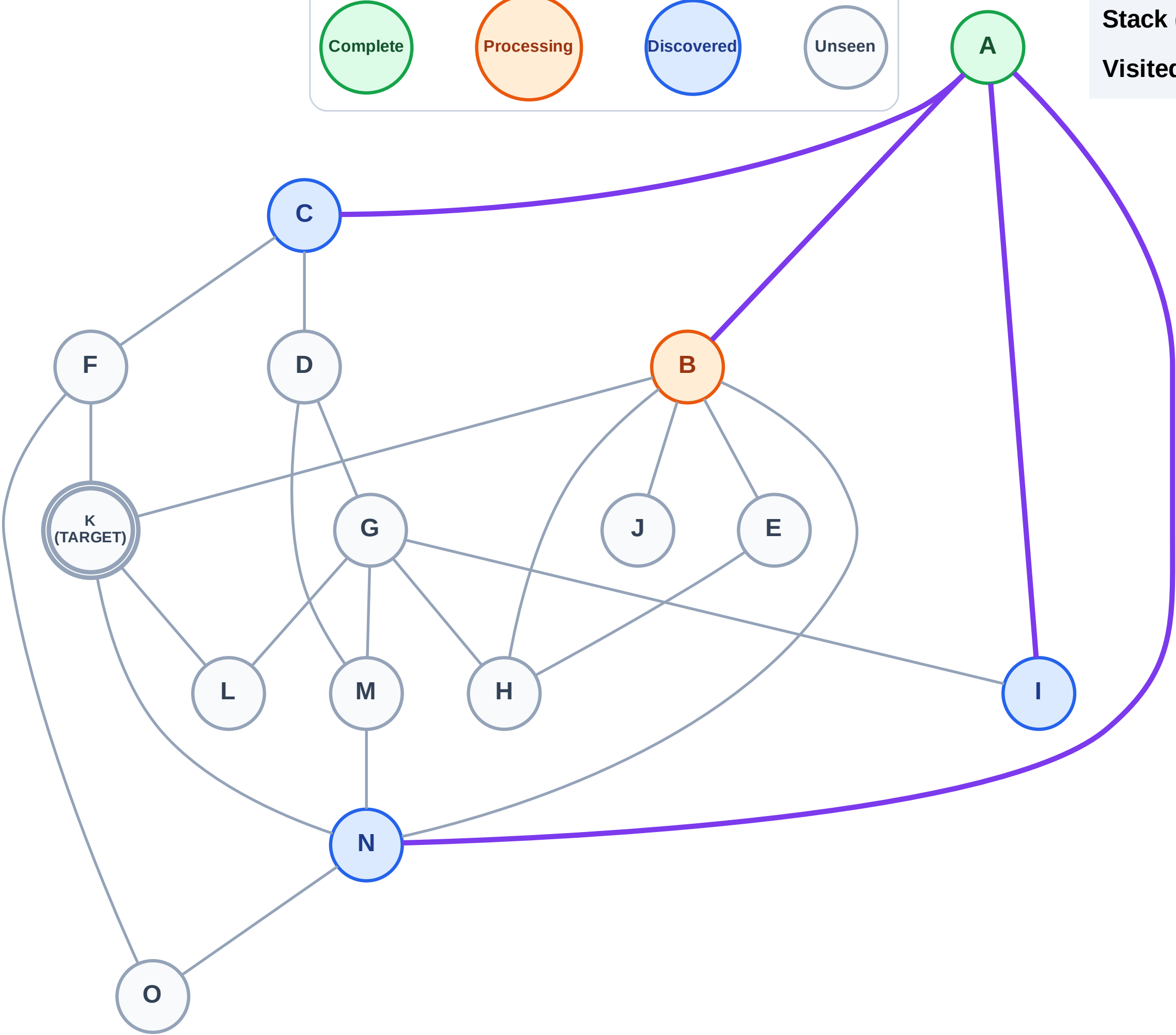
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | C
Visited: A



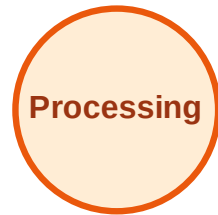
DFS Traversal

Step 5: Discover K, J, H, E from B

Legend



Complete



Processing



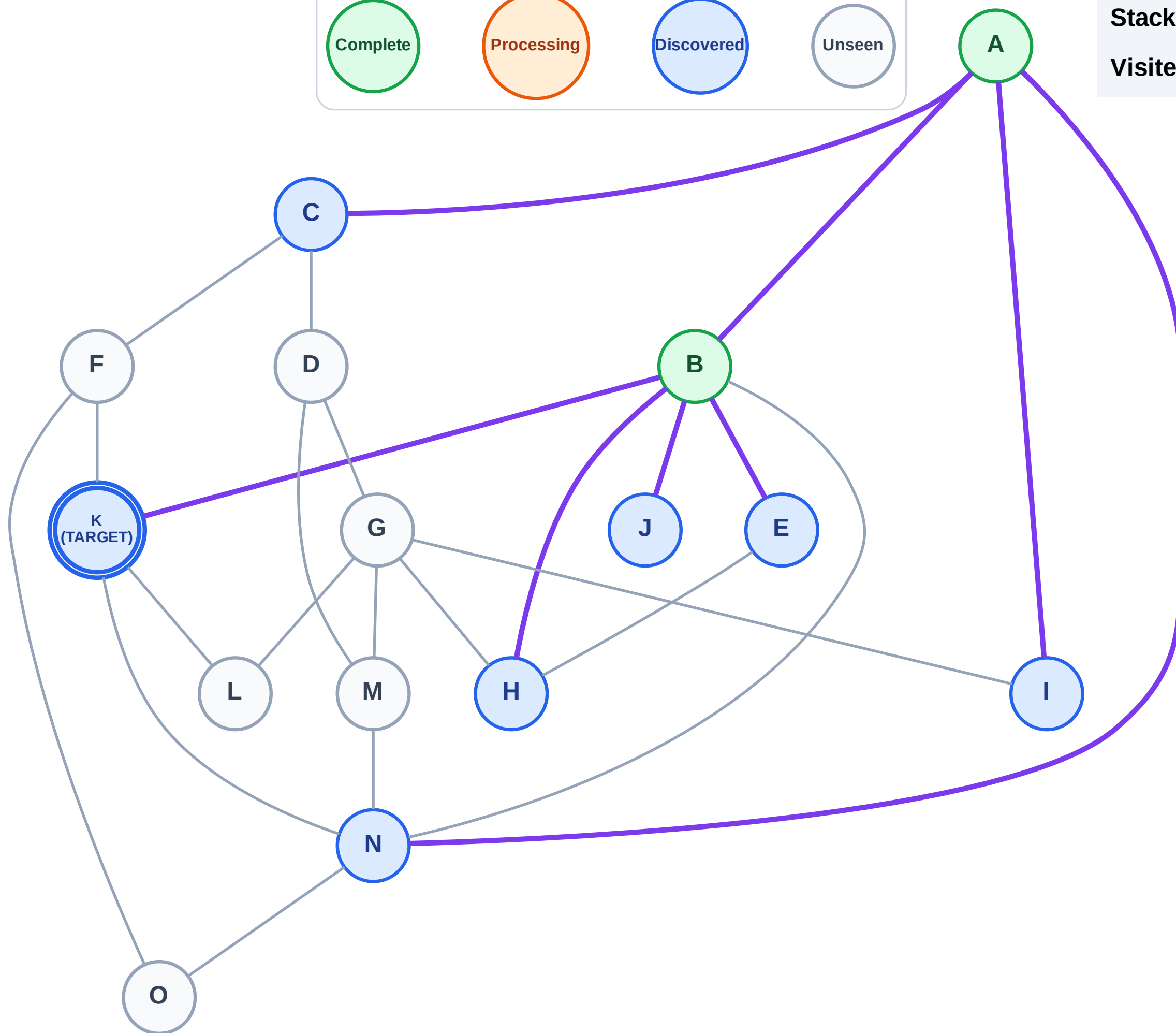
Discovered



Unseen

Stack (top at right): N | I | C | K | J | H | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

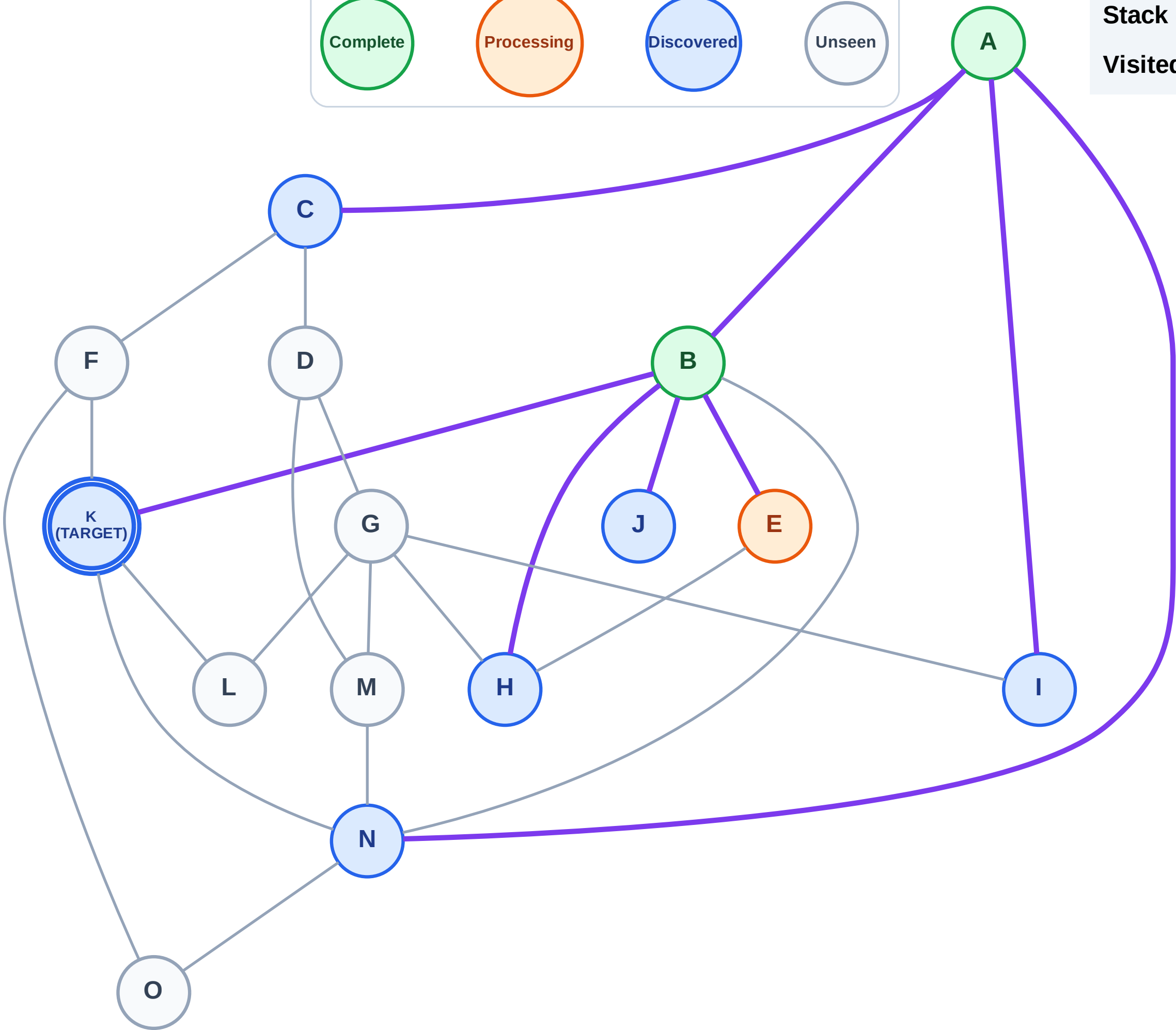
Processing

Discovered

Unseen

Stack (top at right): N | I | C | K | J | H

Visited: A, B



Step 7: No new neighbors from E

Legend



The graph shows a search process starting from node A. The nodes are colored based on their status: Complete (green), Processing (orange), Discovered (blue), and Unseen (grey). The graph shows a path from node A to node K (TARGET). The path is highlighted with thick purple lines. The stack is empty, and the visited set contains nodes A, B, and C.

```

graph TD
    A((A)) --- B((B))
    A --- C((C))
    A --- I((I))
    B --- E((E))
    B --- J((J))
    C --- D((D))
    C --- F((F))
    D --- G((G))
    F --- K((K))
    G --- L((L))
    G --- M((M))
    G --- H((H))
    H --- N((N))
    I --- N((N))
    K --- N((N))
    N --- O((O))
    M --- N((N))
    L --- N((N))
    O --- K((K))
    style A fill:#90EE90,stroke:#333,stroke-width:1px
    style B fill:#90EE90,stroke:#333,stroke-width:1px
    style C fill:#ADD8E6,stroke:#333,stroke-width:1px
    style D fill:#D3D3D3,stroke:#333,stroke-width:1px
    style E fill:#90EE90,stroke:#333,stroke-width:1px
    style F fill:#D3D3D3,stroke:#333,stroke-width:1px
    style G fill:#D3D3D3,stroke:#333,stroke-width:1px
    style H fill:#ADD8E6,stroke:#333,stroke-width:1px
    style I fill:#ADD8E6,stroke:#333,stroke-width:1px
    style J fill:#ADD8E6,stroke:#333,stroke-width:1px
    style K fill:#ADD8E6,stroke:#333,stroke-width:2px
    style L fill:#D3D3D3,stroke:#333,stroke-width:1px
    style M fill:#D3D3D3,stroke:#333,stroke-width:1px
    style N fill:#ADD8E6,stroke:#333,stroke-width:1px
    style O fill:#D3D3D3,stroke:#333,stroke-width:1px
  
```

DFS Traversal

Step 8: Process node H

Legend

Complete

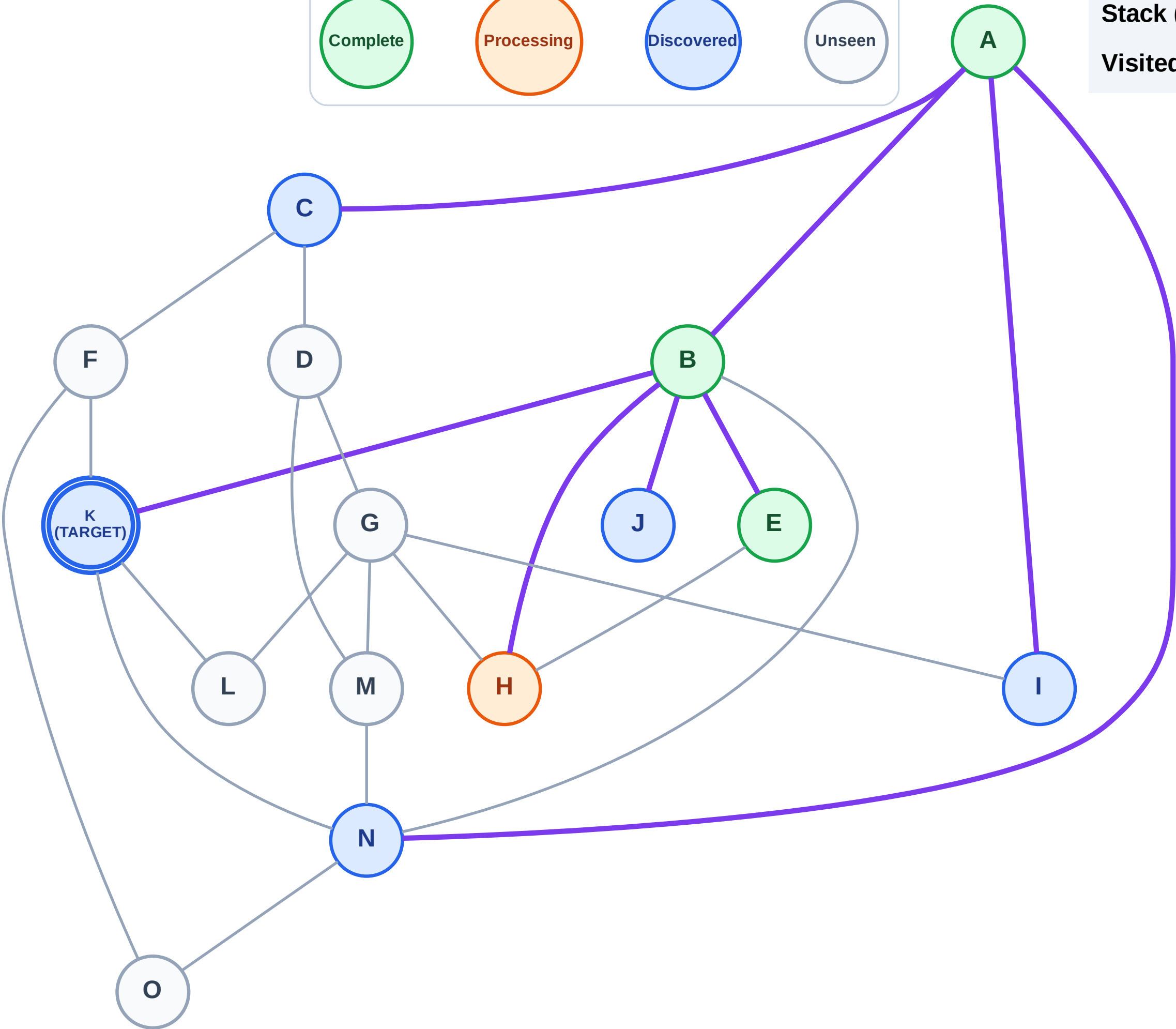
Processing

Discovered

Unseen

Stack (top at right): N | I | C | K | J

Visited: A, B, E



DFS Traversal

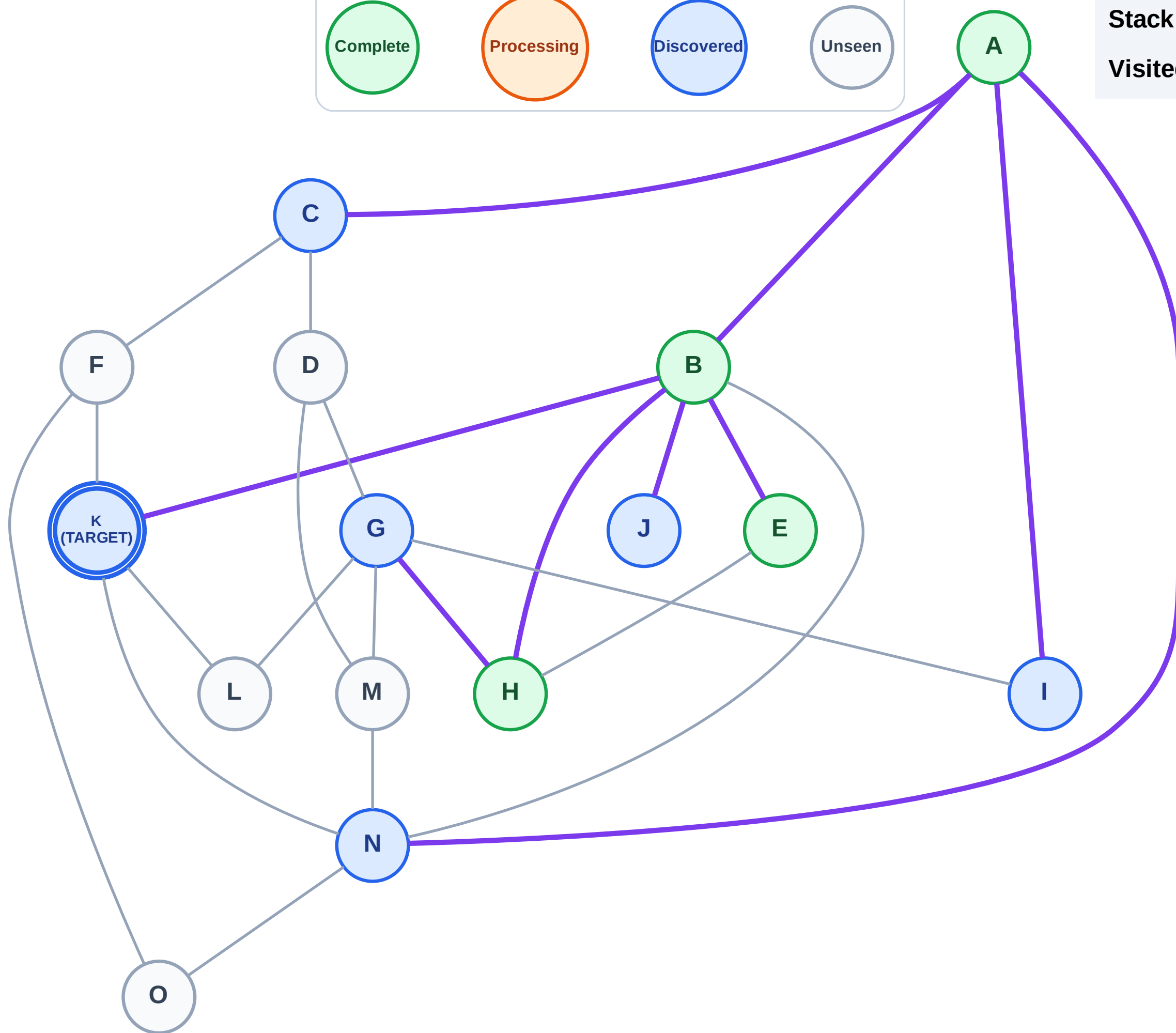
Step 9: Discover G from H

Legend



Stack (top at right): N | I | C | K | J | G

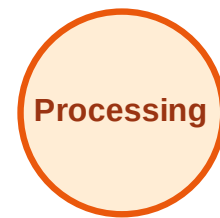
Visited: A, B, E, H



DFS Traversal

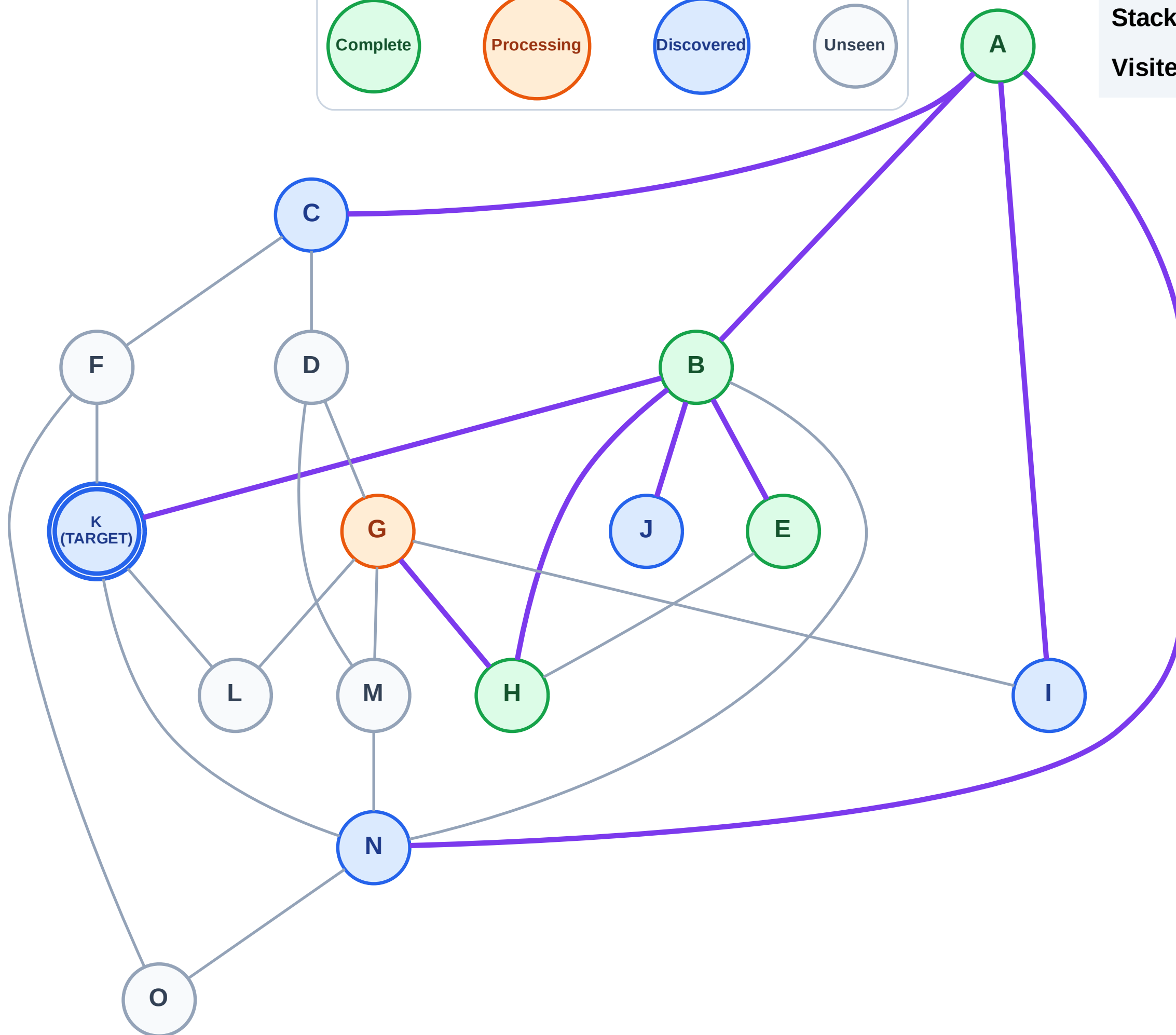
Step 10: Process node G

Legend



Stack (top at right): N | I | C | K | J

Visited: A, B, E, H



Step 11: Discover M, L, D from G

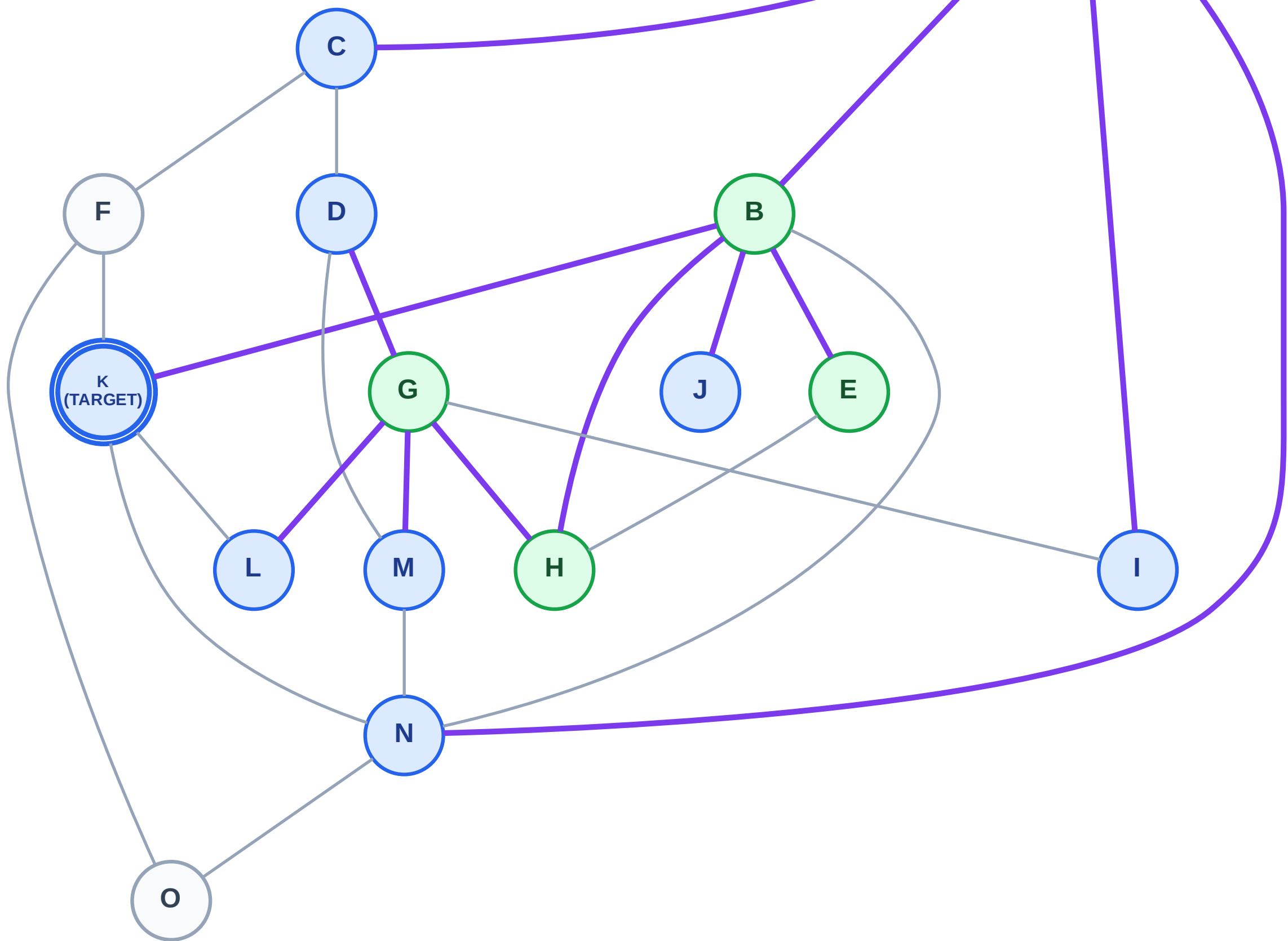
Stack (top at right): N | I | C | K | J | M | L | D

Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle labeled 'Complete', an orange circle labeled 'Processing', a blue circle labeled 'Discovered', and a grey circle labeled 'Unseen'.

- Complete** (Green circle)
- Processing** (Orange circle)
- Discovered** (Blue circle)
- Unseen** (Grey circle)



Step 12: Process node D

Step 12: Process node D



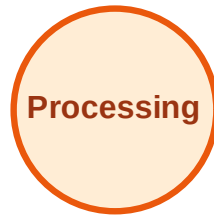
DFS Traversal

Step 13: No new neighbors from D

Legend



Complete



Processing



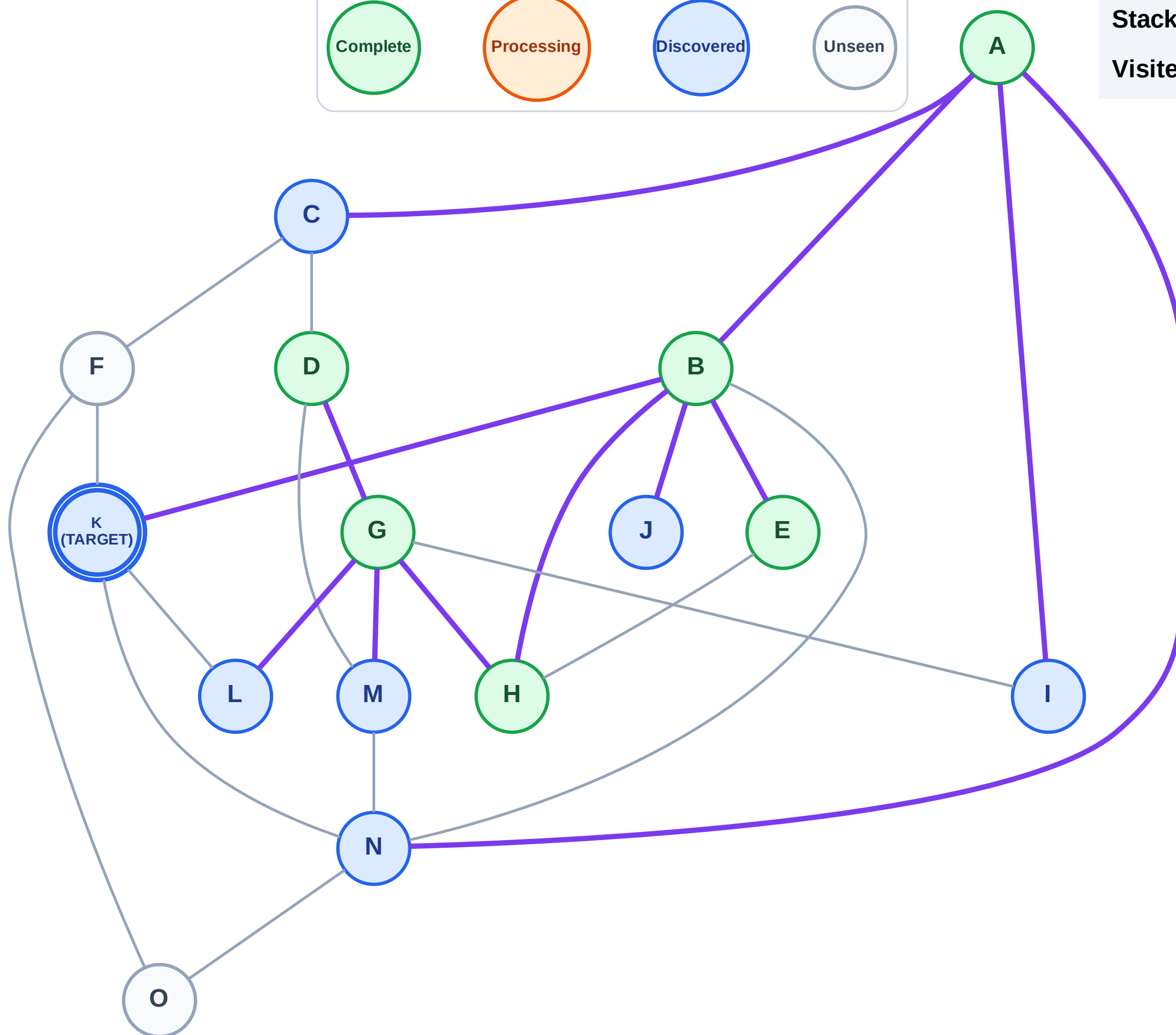
Discovered



Unseen

Stack (top at right): N | I | C | K | J | M | L

Visited: A, B, D, E, G, H



DFS Traversal

Step 14: Process node L

Legend

Complete

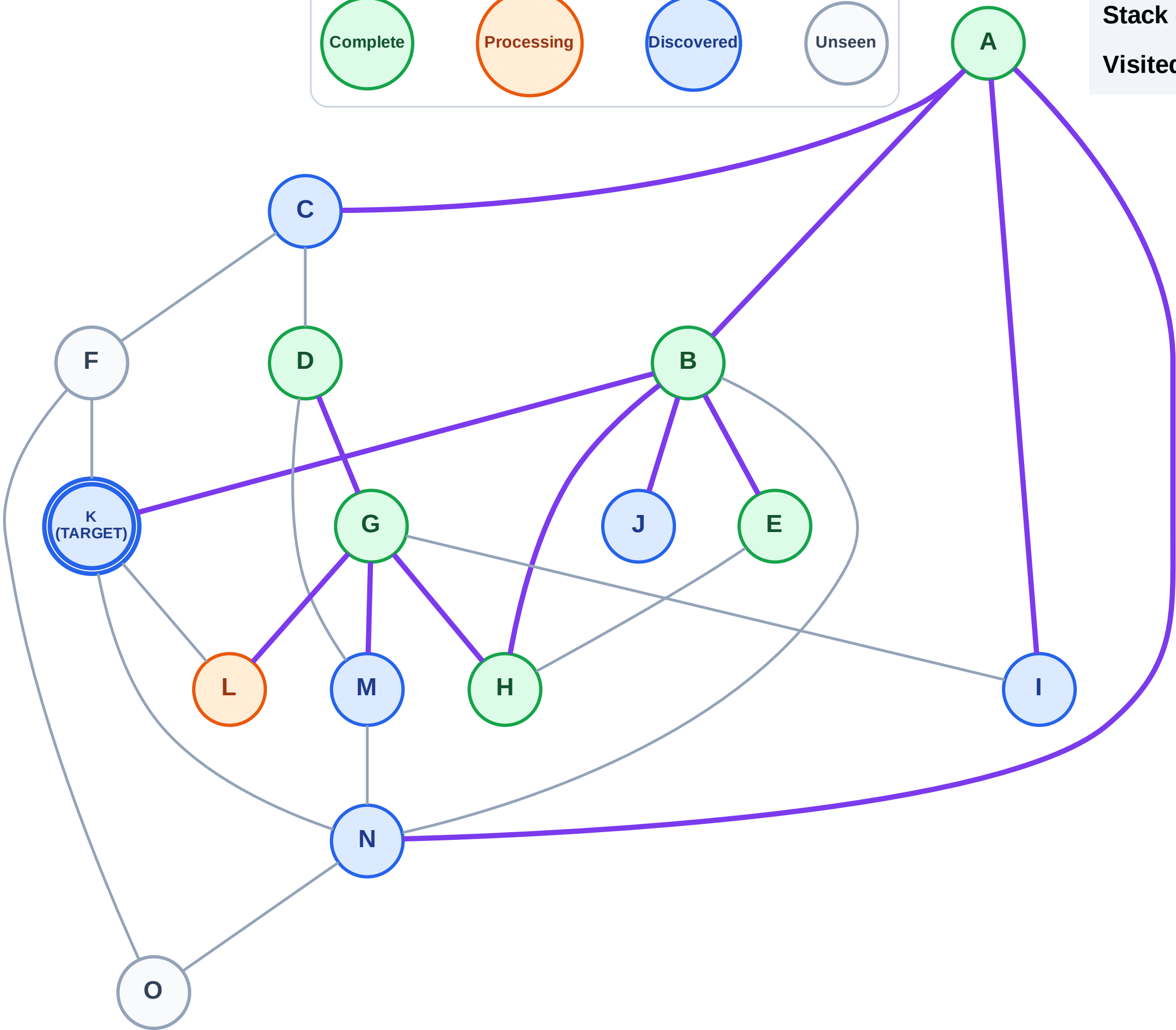
Processing

Discovered

Unseen

Stack (top at right): N | I | C | K | J | M

Visited: A, B, D, E, G, H



DFS Traversal

Step 15: No new neighbors from L

Legend

Complete

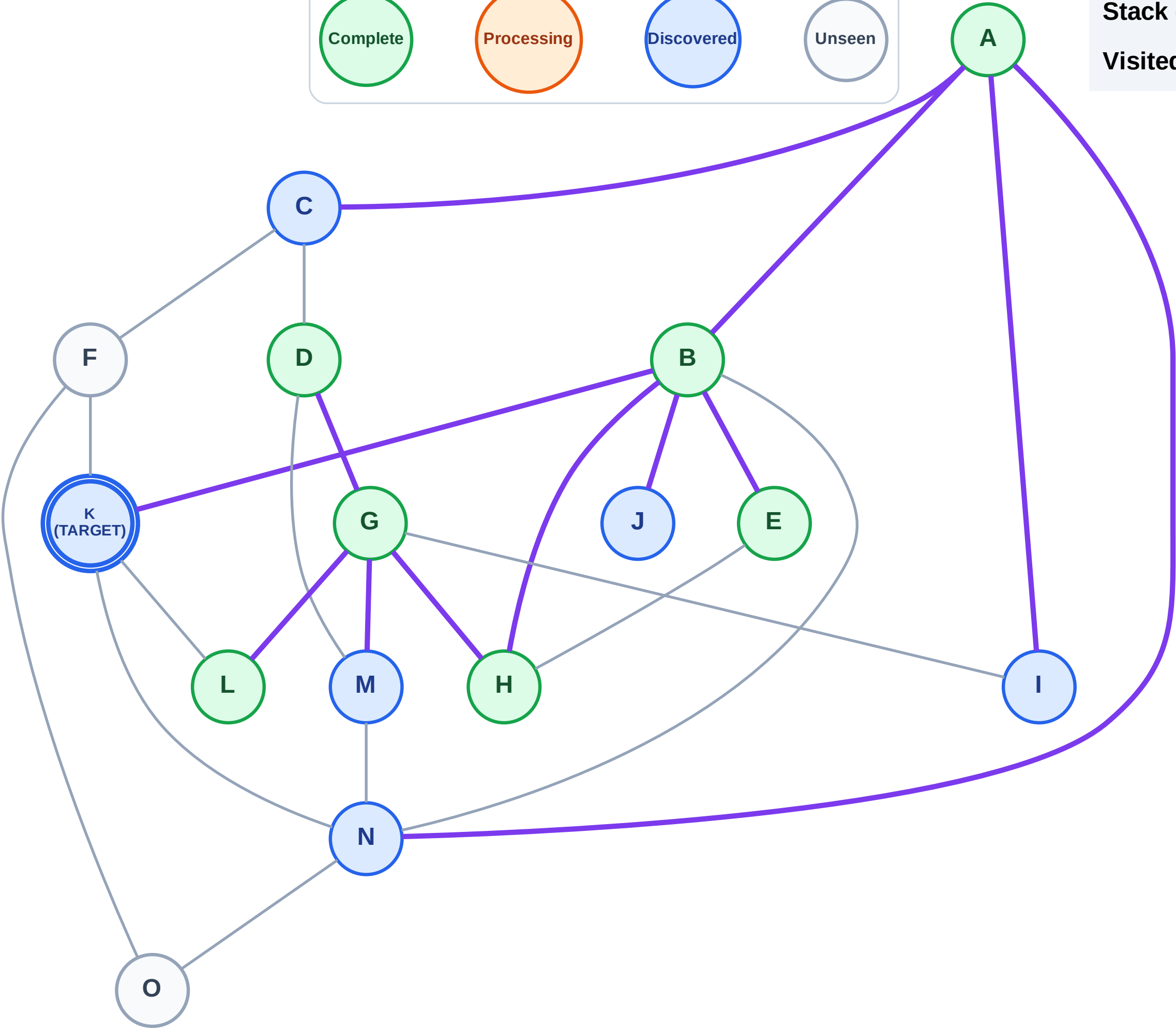
Processing

Discovered

Unseen

Stack (top at right): N | I | C | K | J | M

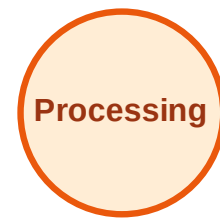
Visited: A, B, D, E, G, H, L



DFS Traversal

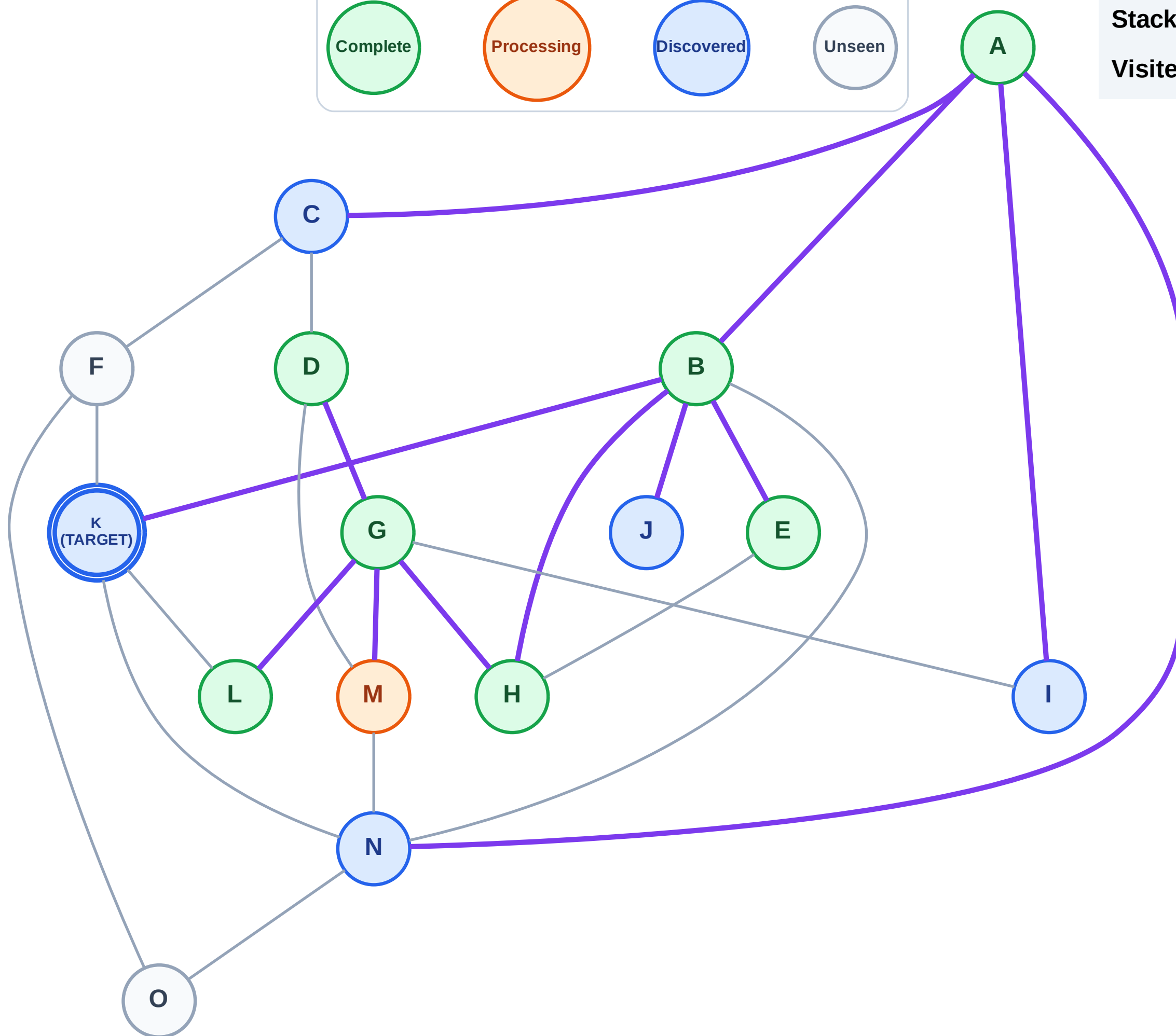
Step 16: Process node M

Legend



Stack (top at right): N | I | C | K | J

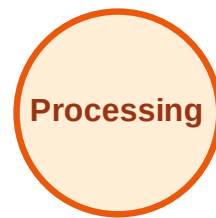
Visited: A, B, D, E, G, H, L



DFS Traversal

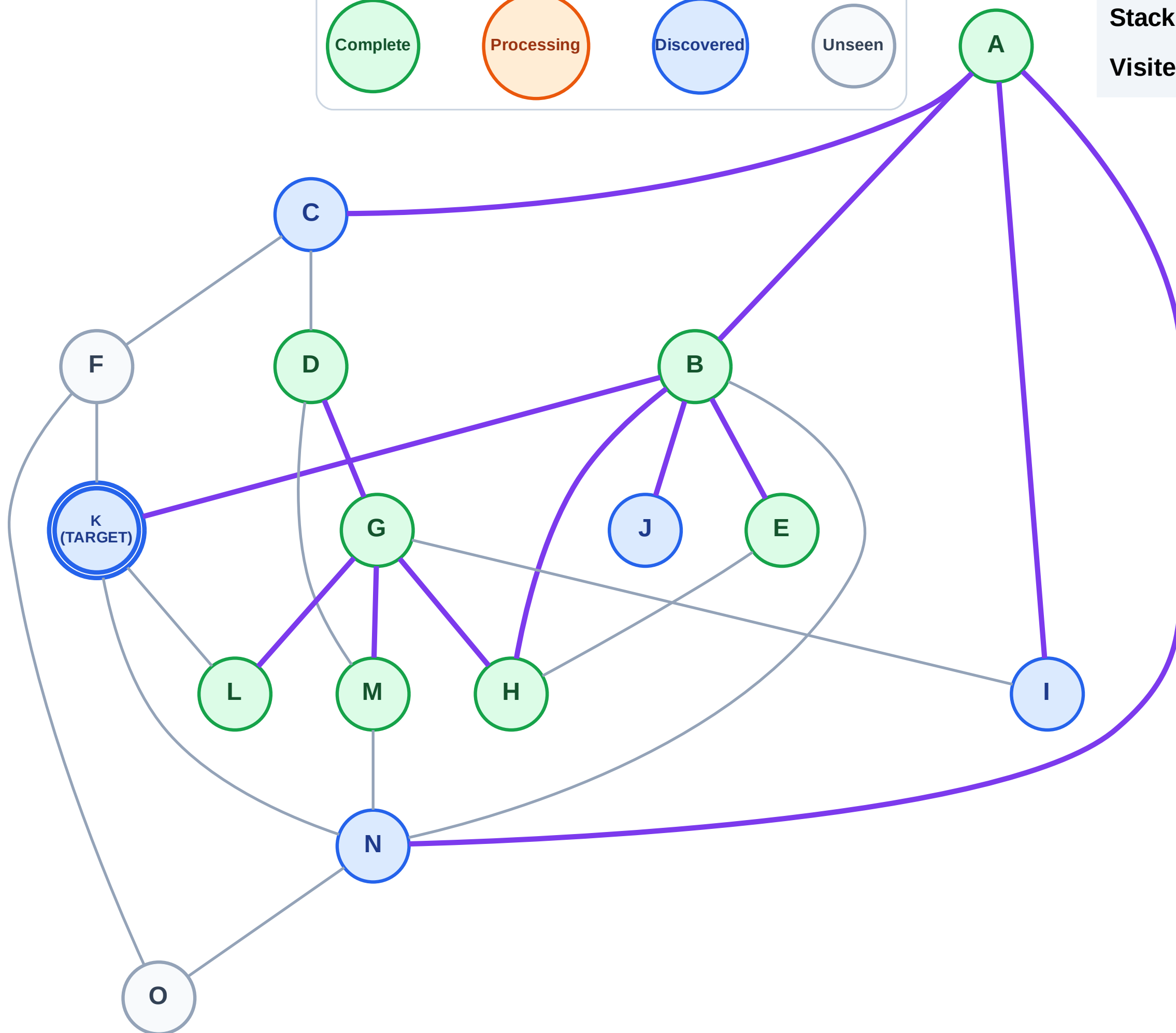
Step 17: No new neighbors from M

Legend



Stack (top at right): N | I | C | K | J

Visited: A, B, D, E, G, H, L, M



DFS Traversal

Step 18: Process node J

Legend

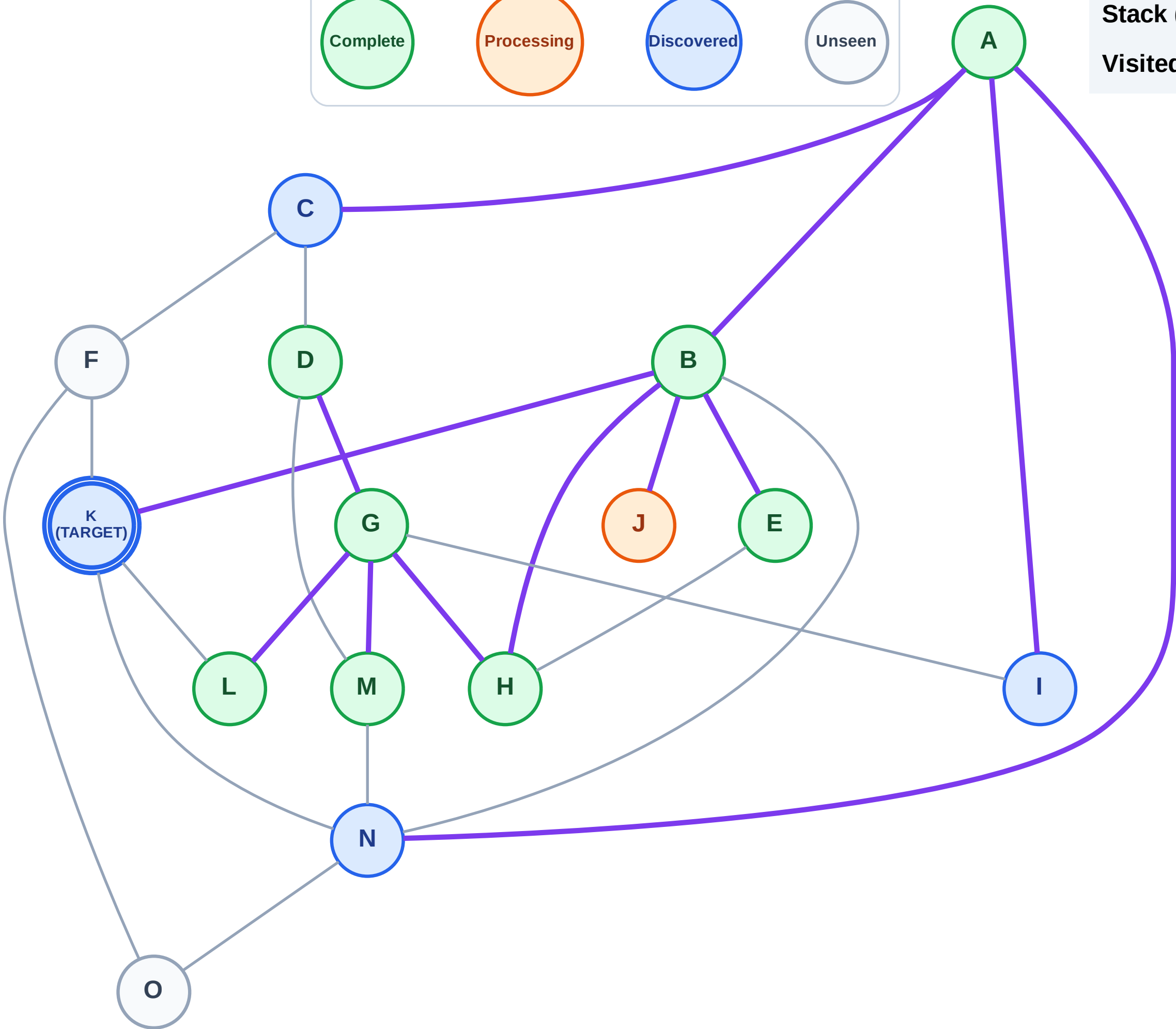
Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | C | K
Visited: A, B, D, E, G, H, L, M



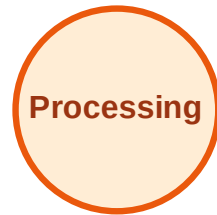
DFS Traversal

Step 19: No new neighbors from J

Legend



Complete



Processing



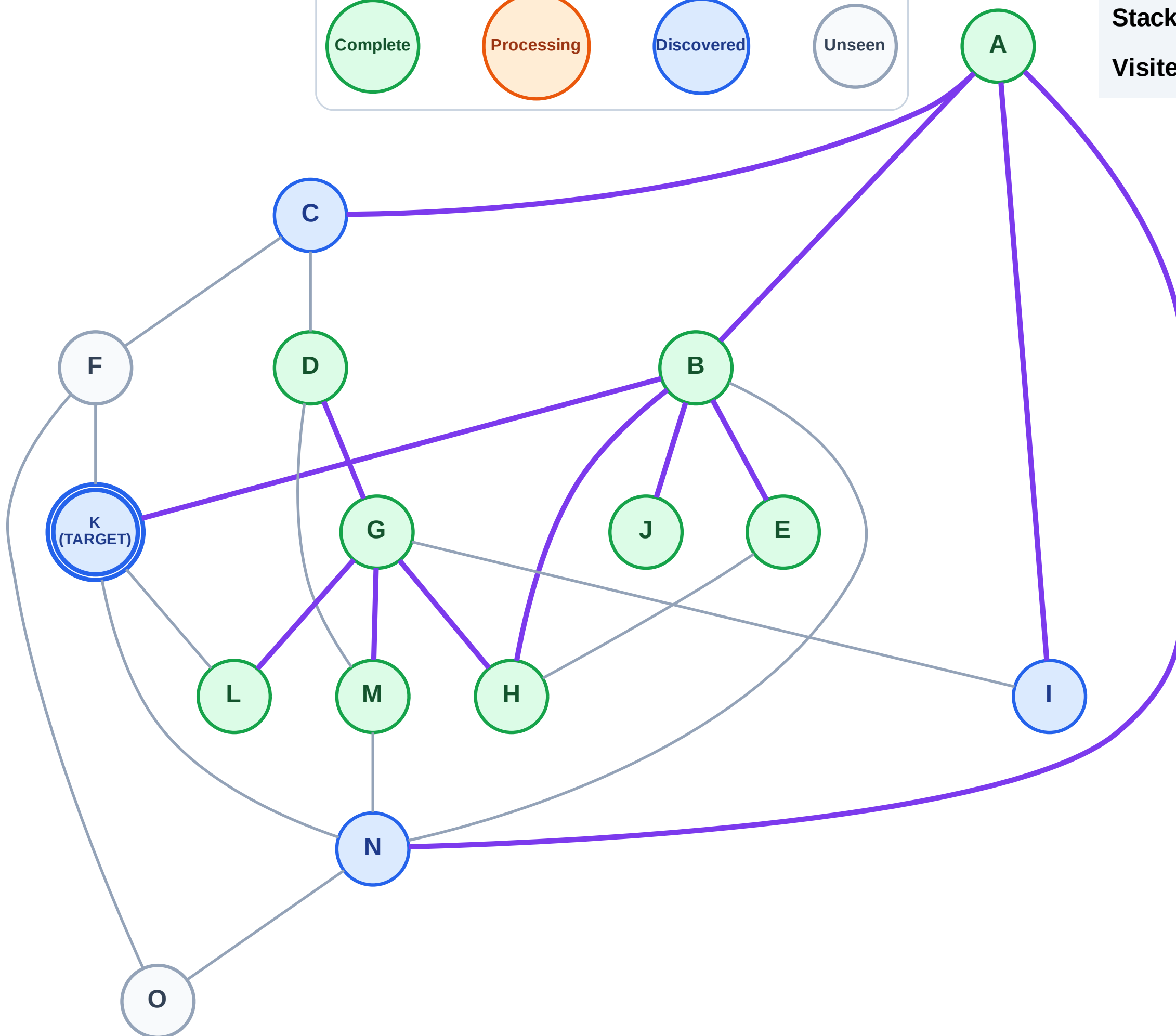
Discovered



Unseen

Stack (top at right): N | I | C | K

Visited: A, B, D, E, G, H, J, L, M



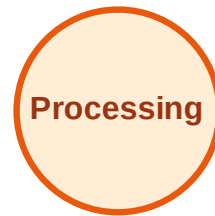
DFS Traversal

Step 20: Process node K

Legend



Complete



Processing

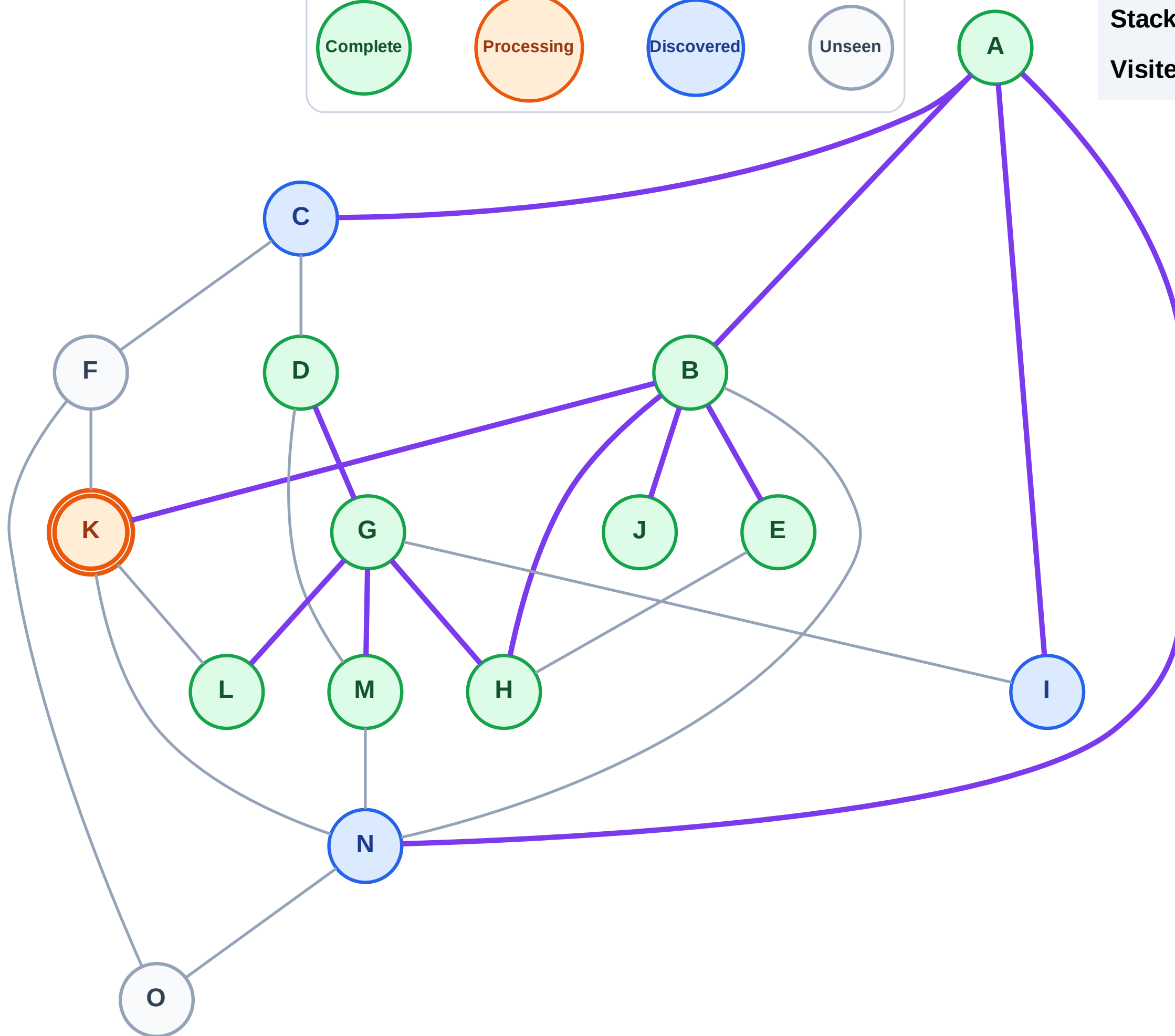


Discovered



Unseen

Stack (top at right): N | I | C
Visited: A, B, D, E, G, H, J, L, M



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | I | C
Visited: A, B, D, E, G, H, J, K, L, M

